

Multinational Firms and Trade in Latin America

Six Stylized Facts on Ownership and Exports

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Abstract

We match the customs export records of nine Latin American countries for 2006–2022 to the ownership data of Orbis and Dun & Bradstreet, so that every exporter is an affiliate of a foreign multinational group, an affiliate of a domestic group, or a local firm. The data cover 165,543 exporters, 5,930 products, 253 destinations and US\$2.4 trillion of exports, and give the country of the ultimate owner for 92% of what foreign affiliates sell abroad. Six facts follow. Multinational affiliates sell 61% of the region's exports, and most of that is foreign owned. Foreign affiliates concentrate in complex and differentiated goods, domestic ones in primary and homogeneous goods. The owners come from few countries: the United States, the United Kingdom and Canada together account for half of foreign-affiliate exports. Groups with more than a hundred affiliates worldwide are 9% of parents and 56% of the value. Where multinationals are present, exports are higher, and the association is about three times larger for foreign affiliates than for domestic ones. Distance holds multinational affiliates back less than it holds back other firms. US groups are a case of their own: they own 10.8% of the region's exports and they are the only large investor that ships a sizeable part of what it produces in the region back home, 22.3% against 0.6% for British parents and close to zero for Swiss and Liechtenstein ones.

Keywords: Multinational Firms, Foreign Direct Investment, Exports, Ownership, Firm-Level Data, Latin America

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1 Introduction

Customs records show what a country sells abroad and to whom. They do not show who owns the seller. For Latin America this is a large blind spot, because much of the export base belongs to firms whose owners sit in another country. Aggregate investment statistics do not close it. They record where capital goes, not which firm ships which product to which market.

This paper closes it for nine countries: Argentina, Chile, Colombia, Costa Rica, the Dominican Republic, El Salvador, Paraguay, Peru and Uruguay. We match their customs export records for 2006–2022 to the ownership records of Orbis and Dun & Bradstreet. The result is a dataset of 5.1 million firm, destination, product and year cells in which every exporter carries a label, foreign affiliate, domestic affiliate or local firm, and in which the sales of foreign affiliates can be traced back to the country of their parent.

Six facts come out of it. First, multinational affiliates sell 61% of the region’s exports, from 47% in Peru to 74% in Colombia, and most of that is foreign owned. Second, foreign affiliates concentrate in complex and differentiated goods and domestic ones in primary and homogeneous goods, with the clearest gradient inside manufacturing. Third, the owners come from few countries: the United States, the United Kingdom and Canada account for half of foreign-affiliate exports. Fourth, a few large groups carry the value, since parents with more than a hundred affiliates worldwide are 9% of parents and 56% of what foreign affiliates sell. Fifth, cells with more multinationals trade more, and the association is about three times larger for foreign affiliates than for domestic ones, including for the local firms selling in the same cell. Sixth, distance reduces the exports of multinational affiliates less than it reduces those of other firms.

The United States runs through all six facts and is the paper’s main story. It is both the largest owner of the region’s exporters and its largest single market, and the two roles almost never meet. US groups own 10.8% of what the nine countries export, 18.4% of those exports go to the United States, and the overlap is 2.5%. What sets US parents apart is that they buy from their own affiliates at all. Of what their Latin American affiliates sell abroad, 22.3% goes to the United States, against 0.6% for British parents, 2.2% for Dutch ones and close to zero for Swiss and Liechtenstein ones. Only Canadian (19.5%) and Brazilian (18.6%) parents come near. The overlap is concentrated in Central America and the Caribbean, where US affiliates move 41% of what El Salvador sends to the United States, 40% of Costa Rica’s and 26% of the Dominican Republic’s, and in manufacturing, where they move 27% of everything the region ships north. In the Southern Cone and the Andes it is small. The nationality of the owner predicts the destination, and it predicts a different one for each owner: British and Liechtenstein groups send mining products to Asia, German and Japanese groups export from Argentina to Brazil.

Contribution to the literature: to be written.

Section 2 reviews the related literature. Section 3 describes the data and how exporters are classified by ownership. Section 4 presents the six facts. Section 5 takes them to the product and sector level. Section 6 sketches the model and Section 7 concludes.

2 Related Literature

To be written. Three strands: firm-level evidence on multinational production and trade; ownership, market power and trade policy; and descriptive evidence on multinational firms in Latin America.

3 Data

3.1 Customs transactions

The trade data are the export records of the customs authorities of nine Latin American countries, obtained by the Inter-American Development Bank under its agreements with the national agencies. A record gives the value exported (FOB, current US dollars) by one identified firm, of one six-digit HS 2007 product, to one destination, in one year. Firms are identified by their national tax number, which is unique within a country, and by the name written on the declaration.

Coverage is the universe of merchandise exports in the years each country made available. Those years differ, from Chile’s 2009–2022 to Paraguay’s 2012–2020, and all nine overlap between 2012 and 2018. Because the panel is unbalanced, every dollar figure in this paper is an annual average, the country’s pooled value divided by the number of years it is observed, while shares are computed on pooled values. Ecuador is in the raw data but excluded throughout. Services do not appear in customs merchandise records and are outside the paper.

3.2 Ownership records

Ownership comes from two commercial databases. Orbis (Bureau van Dijk) gives the ownership links between companies: for each subsidiary, the identity and country of its global ultimate owner at the 25% control threshold. Dun & Bradstreet gives an independent roster with the DUNS number of each establishment and of its global ultimate, together with names, countries and industries. We merge the two into one file in which each firm carries a group identifier, the parent’s country and name, and the number of affiliates the group has worldwide, which is the measure of network size behind Fact 4. The two sources overlap without coinciding: Orbis covers the ownership chain better, Dun & Bradstreet the operating establishments, and some exporters appear in only one.

3.3 Matching exporters to their owners

Customs gives a tax number and a name typed into a declaration. The corporate databases give a registered company name and, for some countries, the same tax number. Matching runs in four steps.

1. **Tax identifier.** Direct match on the national tax number wherever the corporate database records it. This step delivers about 60% of matched export value.
2. **Fuzzy name matching, country by country.** For the rest, company names are compared with a character-level TF–IDF trigram similarity. Character sequences survive the abbreviations that customs declarations make to company names; words do not. A large language model then accepts or rejects each candidate pair.
3. **Fuzzy name matching, pooled.** A second pass over the firms still unmatched, this time against the corporate databases of all countries at once, which catches affiliates registered under a group name rather than a local one.
4. **Manual review.** For the 500 largest unmatched exporters of each country, the match was checked by hand against company websites, corporate registries and public filings. These firms are few and carry a large share of export value, so they drive the aggregate shares.

A parallel step recovers the parent’s country when the ownership record names the parent but not its location, first from the raw Orbis links and then from a model-assisted review of the parent’s name and registered address. After it, the country of the ultimate owner is known for 92.5% of what foreign affiliates export.

3.4 Who counts as a multinational

An exporter found in Orbis or Dun & Bradstreet is an affiliate of a multinational group. We split matched exporters by where the group’s ultimate owner sits: a **domestic** affiliate when the owner is in the exporting country, a **foreign** affiliate otherwise. Matched exporters whose parent country is unknown count as foreign, so the foreign category is what is left of matched firms after the domestic ones. An exporter found in neither database is a **local** firm.

Two consequences matter for reading the exhibits. Because the foreign category is a residual, the aggregate foreign share does not depend on how many parent countries were recovered, which is what Facts 1, 2, 5 and 6 need. And wherever an exhibit splits the foreign bar *by* parent country, the 7.5% of foreign-affiliate value with an unknown parent is allocated across parent countries in proportion to the recorded parents of the same group, so those shares still refer to all foreign affiliates and still add up with the domestic and local ones.

One limit is worth stating. The match runs through ownership links, so a group’s head company cannot be identified as an exporter: every matched exporter in the data has an ultimate owner different from itself, and a Latin American group’s head company exporting from home falls among the unmatched. What the data do show is whether the group is present *at the destination*, either because the affiliate ships to its parent’s country or because the group has another affiliate there. Facts 5 and 6 use that distinction.

3.5 Product classifications and other sources

Products carry the Hausmann–Hidalgo Product Complexity Index, the four technology categories of Lall (2000), the conservative Rauch (1999) split into differentiated, reference-priced and exchange-traded goods, BEC Rev. 4 end use and categories, SITC Rev. 3 divisions, NAICS codes, upstreamness, and the import-demand elasticities of Fontagné, Guimbard and Orefice (2022), all mapped to HS 2007 six-digit lines. Sectors are defined on HS chapters: agriculture 01–24, mining and fuels 25–27 and 71, manufacturing 28–97 excluding 71. Distance, contiguity, common language and trade agreements come from the CEPII gravity database.

3.6 The working sample

Table 1 describes the sample country by country and Table 2 summarizes it. The nine countries contribute 5.1 million firm, destination, product and year records, 165,543 exporters of which 31,350 are matched to a corporate database, 5,930 HS6 lines, 253 destinations and US\$2,394 billion of pooled export value. The matched exporters belong to 17,094 groups whose owners sit in 165 countries. Regressions use plain logarithms, so a cell with a zero count drops out of that column, and cluster standard errors at the origin–destination level.

Table 1: The working sample: coverage by exporting country

Origin	Years	Exporters	Matched	HS6 lines	Destinations	Records	Value (\$bn/yr)
ARG	2011–2019	22,980	6,687	4,657	202	1,105,127	66.4
COL	2010–2021	47,711	7,711	4,850	223	1,164,935	44.0
PER	2010–2019	27,286	4,696	4,600	215	845,003	41.4
CHL	2009–2022	17,430	5,776	4,673	205	589,227	34.2
CRI	2010–2019	14,816	1,387	4,452	197	530,375	10.5
DOM	2012–2019	15,832	1,671	3,970	206	329,133	8.6
PRY	2012–2020	3,220	999	2,801	168	60,188	7.3
URY	2010–2019	5,828	1,465	3,611	208	121,452	7.0
SLV	2006–2018	10,440	958	4,310	183	362,774	5.0
All nine	2006–2022	165,543	31,350	5,930	253	5,108,214	224.5

One record = firm $\times$ destination $\times$ HS6 product $\times$ year. Exporters are counted as distinct tax identifiers within an origin; matched exporters are those found in Orbis or Dun & Bradstreet after the tax-identifier match, the fuzzy name match and the manual review of the largest exporters. Value is the annual average: the origin's pooled export value divided by the number of years it is observed. Ecuador is in the raw data but is excluded throughout.

Table 2: The working sample in figures

The working sample	
Exporting countries	9
Years	2006–2022
Records (firm $\times$ destination $\times$ HS6 $\times$ year)	5,108,214
Exporters	165,543
matched to Orbis / Dun & Bradstreet	31,350
HS6 product lines	5,930
Destination markets	253
Parent countries of foreign multinationals	165
Multinational groups	17,094
Export value, pooled (\$bn)	2,394
foreign multinationals	46.3%
domestic multinationals	14.1%
local firms	39.5%
Foreign-MNE value with a recorded parent country	92.5%

4 Six Stylized Facts on Multinational Firms and Trade

4.1 Fact 1: Multinational corporations account for a large share of export values across countries

Figure 1 gives, for each country, the share of export value sold by multinational affiliates, split into foreign and domestic owners. Multinationals sell between 47% (Peru) and 74% (Colombia) of exports, and 61% for the nine pooled. Foreign owners dominate: 46% of total exports against 14% for domestic multinationals. Colombia is the exception, where the state oil company and its subsidiaries push the domestic share above the foreign one, and Chile is the near exception, where copper and pulp do the same on a smaller scale. A large domestic multinational presence is a natural-resource story in two countries.

The United States is both the largest owner and the largest market, so it is worth separating the two roles at the start. Of the region's exports, 18.4% go to the United States and 10.8% are sold by affiliates of US groups. Only 2.5% is both, and US affiliates move 13.6% of everything the nine countries ship to the United States. The overlap sits in Central America and the Caribbean, at 19.0% of El Salvador's exports, 15.1% of Costa Rica's and 12.5% of the Dominican Republic's, and is small everywhere else.

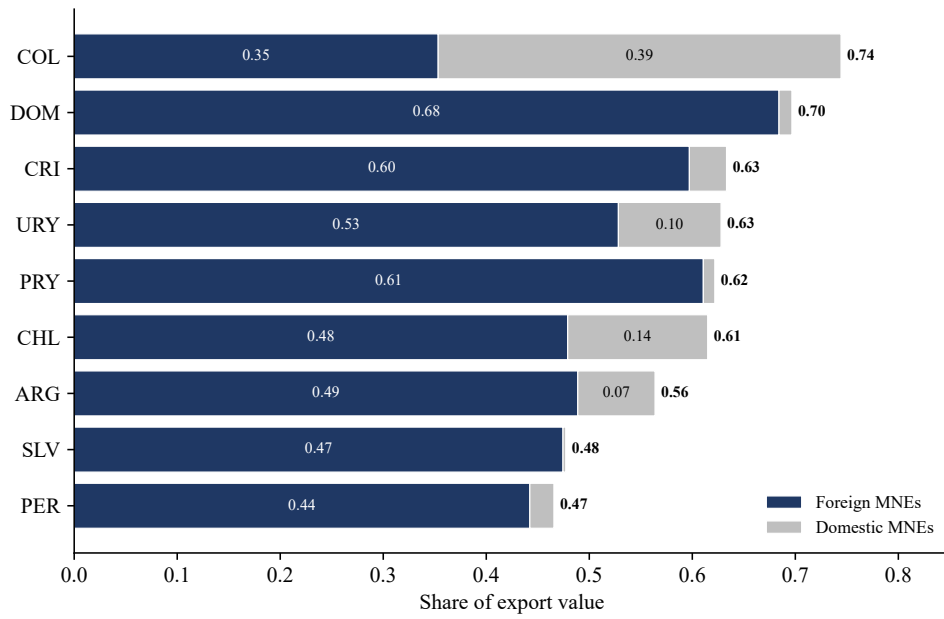


Figure 1: MNE share of export value by origin: foreign and domestic multinationals.

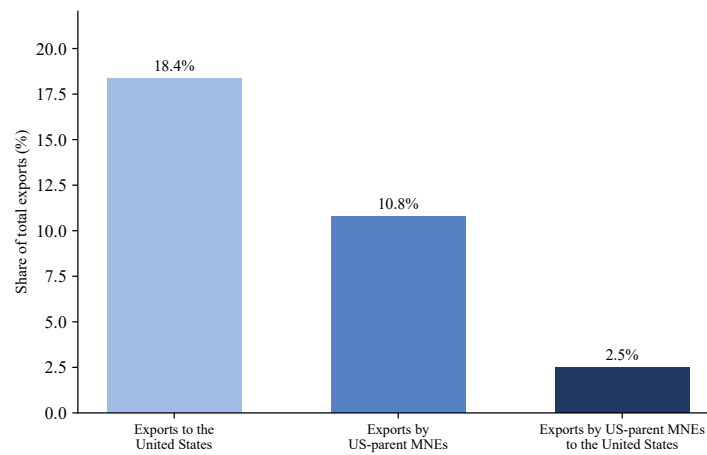


Figure 2: Exports of the nine origins, as shares of total exports: going to the United States, carried by US-parent MNEs, and both.

Note: Same denominator for the three bars: total export value of the nine origins, all firms and destinations. The second bar is the USA's share of foreign-MNE value (23.3 %, Fact 3) times the foreign-MNE share of exports (46.3 %, Figure 1); the third bar is the second times the share of US-parent exports that goes to the United States (22.3 %, the diagonal of Table 8).

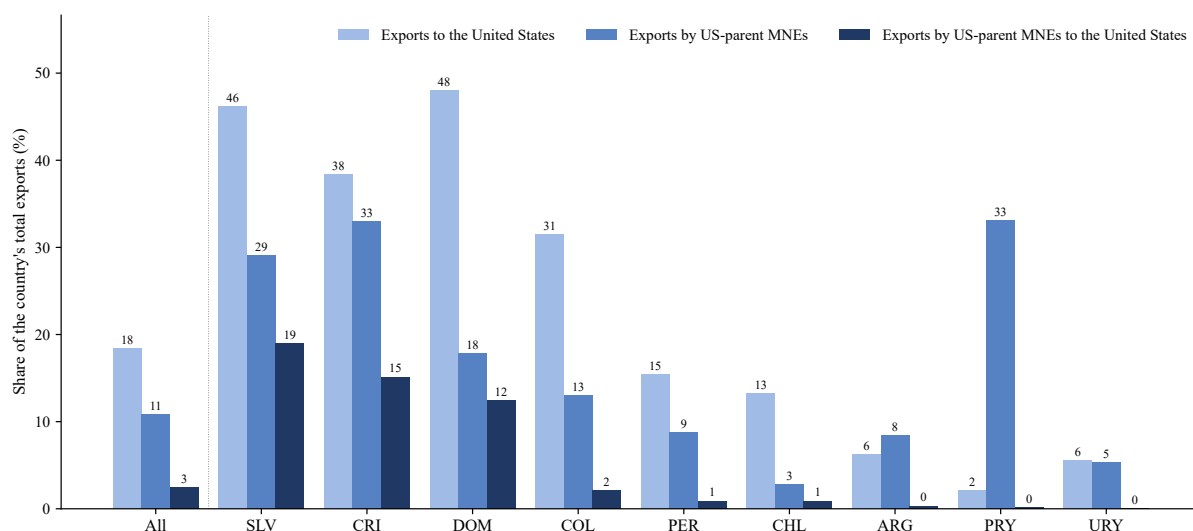


Figure 3: The same three shares by origin (denominator: the origin's total exports).

Table 3: Exports going to the United States, carried by US-parent MNEs, and both, as shares of total exports; and the US-parent share of the exports going to the United States

Origin	(1) % of exports going to the USA	(2) % of exports carried by US-parent MNEs	(3) % of exports carried by US-parent MNEs and going to the USA	(4) = (3)/(1): US-parent share of the exports going to the USA (%)	Total exports (\$bn/yr)
All	18.4	10.8	2.5	13.6	224.5
SLV	46.1	29.1	19.0	41.3	5.0
CRI	38.3	33.0	15.1	39.5	10.5
DOM	48.0	17.8	12.5	26.0	8.6
COL	31.5	13.1	2.1	6.8	44.0
PER	15.4	8.8	0.9	6.0	41.4
CHL	13.3	2.8	0.9	6.9	34.2
ARG	6.3	8.4	0.3	5.5	66.4
PRY	2.1	33.1	0.2	8.1	7.3
URY	5.5	5.3	0.0	0.4	7.0

Columns (1)–(3) have the same denominator, the origin's total exports (all firms, all destinations): (1) value shipped to the United States by any firm; (2) value exported by multinationals whose ultimate parent is in the United States, to any destination; (3) the intersection. Column (4) changes the denominator to the exports that go to the United States and is the ratio (3)/(1); it is the US-parent bar of the 'who carries the exports to the USA' figure. Origins sorted by column (3). Foreign MNEs whose parent country is not recorded (7.5% of foreign-MNE export value) are allocated to parent countries in proportion to the recorded parents of the same group, so every share by parent country refers to all foreign MNEs and adds up with the domestic and local shares. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

4.2 Fact 2: Foreign multinationals specialize in complex products, domestic ones in primary goods

Figure 4 sorts products by sophistication. The foreign share climbs from 0.46 in the least complex quintile of the Product Complexity Index to 0.65 in the most complex one, and is highest in high- and medium-technology manufactures. Domestic multinationals run the other way, concentrated in primary

and resource-based goods. Figure 5 shows that affiliates of OECD-headquartered groups carry the gradient.

The same ordering shows up in classifications that use no complexity index. Foreign affiliates hold 56% of differentiated products and 55% of reference-priced ones against 34% of exchange-traded goods, where the domestic share is four times its average (Table 4 and Figure 6). By end use they hold 65% of capital goods against 48% of intermediates and 44% of consumption goods (Table 5). Table 6 runs the same comparison in origin, destination, product and year cells. The pattern is sharpest inside manufacturing (Figure 7). It is noisy inside agriculture and absent inside mining, where too few products exist for the quintiles to order anything.

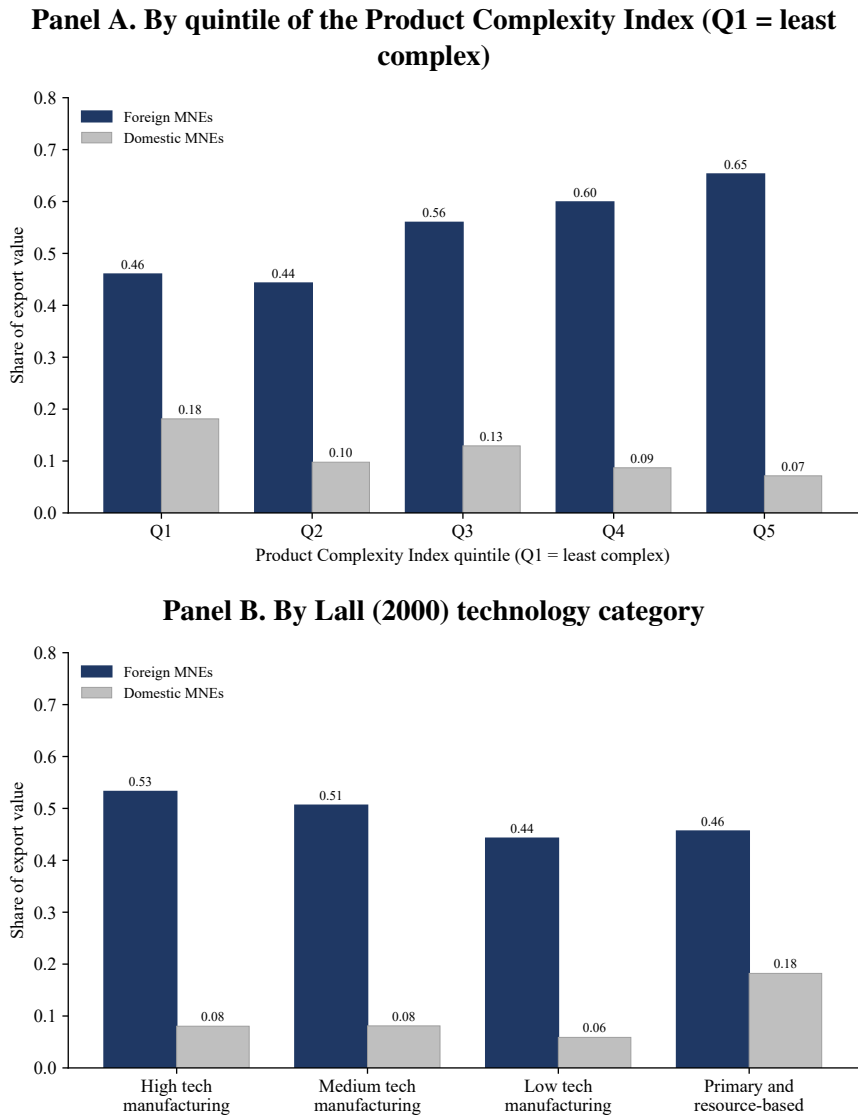


Figure 4: MNE share of export value by product sophistication: foreign and domestic multinationals.

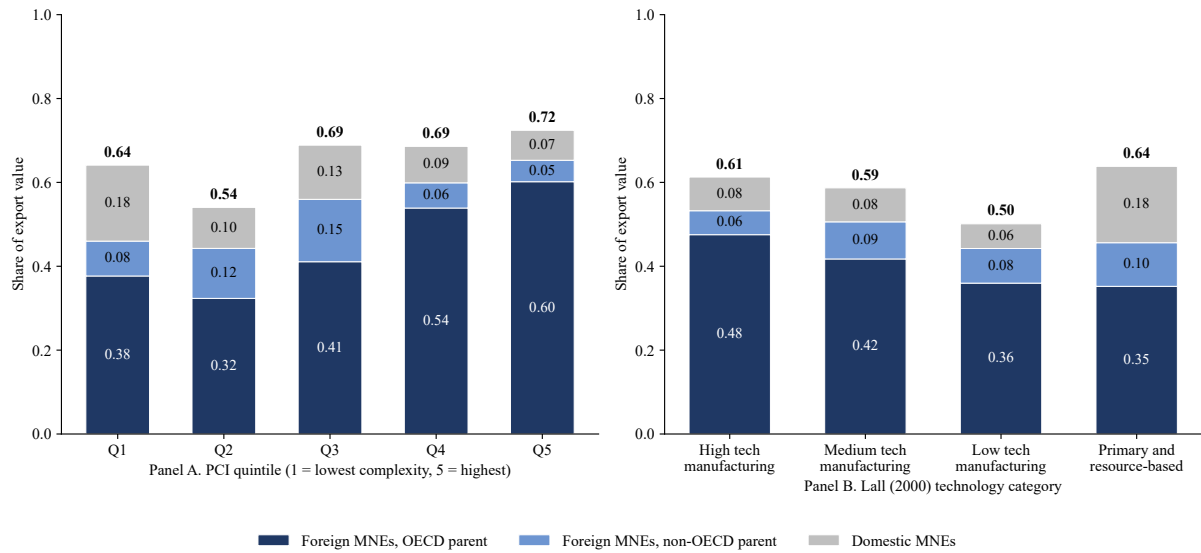


Figure 5: MNE share by PCI quintile (Panel A) and Lall category (Panel B), foreign bar split into OECD-parent and non-OECD-parent multinationals.

Note: OECD = the 38 member countries; matched firms with no recorded parent country are counted with the non-OECD parents.

Table 4: MNE shares by Rauch (1999) class: differentiated, reference-priced, homogeneous

Rauch (1999) class	Foreign	Domestic	MNE total	N HS6	Value (\$bn/yr)
Differentiated	0.563	0.058	0.621	2,712	44.3
Reference-priced	0.549	0.082	0.631	1,636	63.7
Homogeneous (exchange)	0.343	0.228	0.571	319	97.6

Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

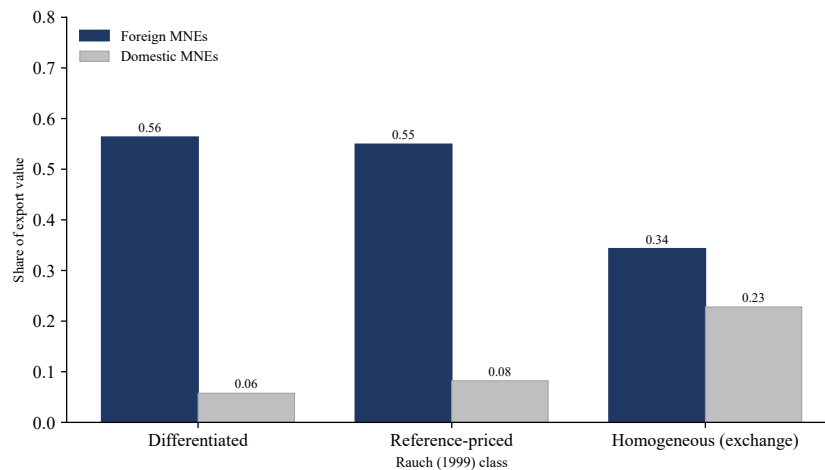


Figure 6: Foreign vs domestic MNE shares of export value by Rauch (1999) class.

Table 5: MNE shares by BEC end use: intermediate, consumption and capital goods

BEC end use	Foreign	Domestic	MNE total	N HS6	Value (\$bn/yr)
Intermediate	0.476	0.155	0.631	3,090	155.0
Consumption	0.438	0.082	0.520	1,201	46.1
Capital	0.647	0.056	0.703	659	9.3

BEC Rev. 4 end use in three classes: intermediate = industrial supplies, fuels (incl. motor spirit and goods n.e.s., which are fuels in these data), parts and food for industry; consumption = food for households, passenger cars and consumer goods; capital = capital goods and industrial transport equipment. HS6 lines without a BEC correspondence are excluded. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 6: Product sophistication and MNE shares: PCI, Lall categories and Rauch classes, each in a separate regression

	PCI		Lall		Rauch	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: MNE_{total} share</i>						
Complexity (PCI)	-0.0004 (0.0050)	0.0052 (0.0045)				
Lall: high tech (vs primary and resource-based)			-0.0288** (0.0142)	-0.0347** (0.0142)		
Lall: medium tech			-0.0256*** (0.0081)	-0.0111 (0.0074)		
Lall: low tech			-0.0801*** (0.0165)	-0.0797*** (0.0171)		
Rauch: differentiated (vs homogeneous)					0.0819*** (0.0112)	0.0880*** (0.0104)
Rauch: reference-priced					0.1038*** (0.0103)	0.1132*** (0.0104)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
<i>Panel B: MNE_{ext} share</i>						
Complexity (PCI)	0.0385*** (0.0052)	0.0280*** (0.0050)				
Lall: high tech (vs primary and resource-based)			0.0330** (0.0135)	0.0140 (0.0129)		
Lall: medium tech			0.0400*** (0.0080)	0.0517*** (0.0071)		
Lall: low tech			-0.0000 (0.0118)	-0.0050 (0.0087)		
Rauch: differentiated (vs homogeneous)					0.2088*** (0.0082)	0.1914*** (0.0079)
Rauch: reference-priced					0.1783*** (0.0099)	0.1659*** (0.0099)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
<i>Panel C: MNE_{dom} share</i>						
Complexity (PCI)	-0.0389*** (0.0058)	-0.0227*** (0.0045)				
Lall: high tech (vs primary and resource-based)			-0.0618*** (0.0143)	-0.0487*** (0.0137)		
Lall: medium tech			-0.0657*** (0.0087)	-0.0628*** (0.0075)		
Lall: low tech			-0.0801*** (0.0148)	-0.0747*** (0.0155)		
Rauch: differentiated (vs homogeneous)					-0.1270*** (0.0101)	-0.1034*** (0.0084)
Rauch: reference-priced					-0.0745*** (0.0086)	-0.0527*** (0.0075)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
Origin × year and destination × year FE	✓		✓		✓	
Origin × dest. × year FE		✓		✓		✓

One observation = origin × destination × HS6 product × year; dep. var. MNE share of the export value of the product from the origin to the destination in the year; each measure entered in its own regression; weighted by that export value; robust (HC1) SE. PCI = Hausmann–Hidalgo Product Complexity Index (continuous); Lall (2000) in the four categories of Figure 3, base = primary and resource-based; Rauch (1999) in three classes, base = homogeneous (exchange-traded). Cells whose HS6 line lacks the measure are dropped from that column. *** p<0.01, ** p<0.05, * p<0.1

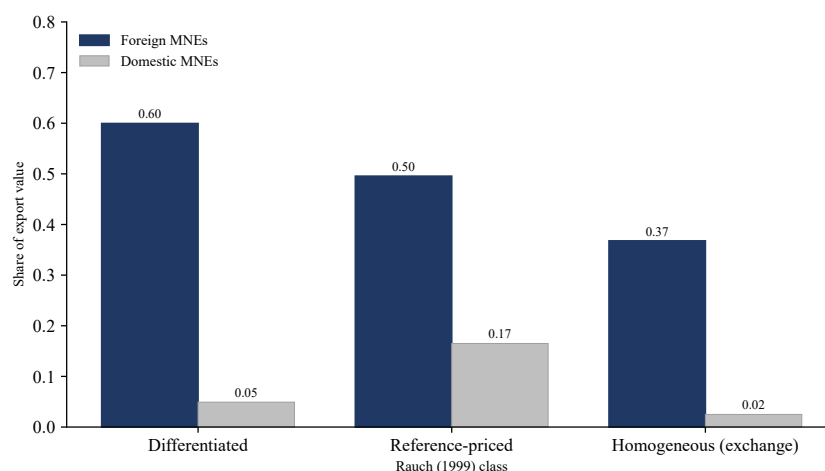


Figure 7: Manufacturing: foreign vs domestic MNE shares by Rauch (1999) class.

4.3 Fact 3: Multinational corporations from a small set of countries dominate exports

Figure 8 ranks the owners of foreign affiliates by their share of foreign-affiliate export value. The United States (23%), the United Kingdom (19%) and Canada (8%) account for half of it, the ten largest owners for more than three quarters, and the remaining 150 countries for the rest.

Where the goods go depends on who owns the seller. Tables 7 and 8 give the destination mix of the ten largest owners, the second using the owners' own countries as destinations, so its diagonal is the share each group of affiliates ships home. Three routes stand out. British and Liechtenstein groups, concentrated in mining, sell to Asia. German and Japanese groups export from Argentina to Brazil, treating the region as a platform for Mercosur. And US groups ship home: 22.3% of what their affiliates sell, against 0.6% for British owners and 2.2% for Dutch ones, with only Canada (19.5%) and Brazil (18.6%) close behind. Figure 9 looks at the same pattern from the destination side and asks, for each large market, how much of what arrives is carried by affiliates of groups headquartered there.

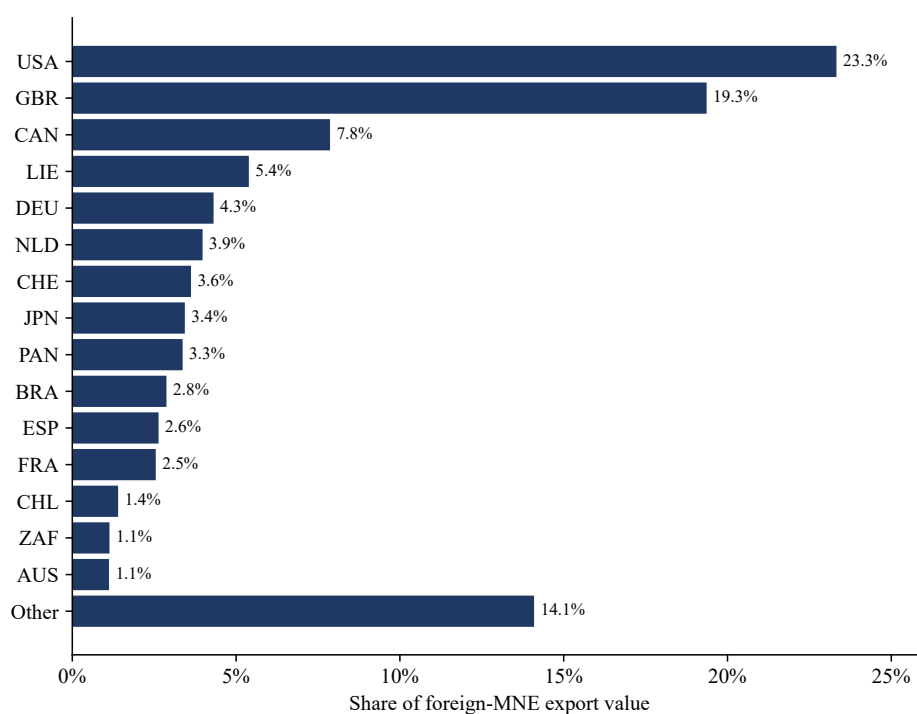


Figure 8: Foreign-MNE export value by the parent's country: top 15 and other.

Note: Share of all foreign-MNE export value. Foreign MNEs export 46.3 % of total exports (Figure 1), so the USA's 23.3 % of foreign-MNE value is $0.233 \times 46.3 = 10.8\%$ of total exports, the US-parent share of Table 3.

Table 7: Top-10 parents $\times$ top-10 destinations: destination mix of each parent's exports (row %)

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations	Total
USA	10.1	22.3	7.8	3.8	4.0	2.9	2.1	0.5	3.5	4.9	38.2	100.0
GBR	36.5	5.6	3.9	7.1	3.0	3.4	6.5	3.3	3.2	0.2	27.3	100.0
CAN	12.8	6.6	4.2	5.3	3.1	0.5	1.9	19.5	0.9	0.2	45.0	100.0
LIE	24.9	3.7	3.7	18.7	3.5	2.5	5.4	0.0	1.8	0.0	35.6	100.0
DEU	1.0	14.8	31.9	1.1	1.7	3.1	0.6	1.1	1.0	0.9	42.8	100.0
NLD	9.2	3.8	26.9	0.4	1.0	3.6	1.1	0.2	2.2	3.1	48.4	100.0
CHE	6.2	5.0	6.8	0.5	2.9	3.5	0.8	0.4	2.0	0.5	71.4	100.0
JPN	2.4	8.7	45.0	8.9	0.4	1.4	0.4	0.5	0.9	1.9	29.5	100.0
PAN	35.0	9.5	1.2	1.8	1.7	1.1	6.2	0.6	2.0	0.2	40.7	100.0
BRA	12.9	14.1	18.6	1.6	0.6	3.1	0.6	0.6	2.1	2.3	43.4	100.0
Other parents	9.5	19.5	11.7	2.2	3.7	4.0	1.6	1.6	2.1	3.7	40.3	100.0

Row percentages: destination mix of each row group. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

Table 8: Top-10 parents × the same ten countries as destinations (row %): the diagonal is the share of each parent's exports shipped to the parent's own country

Parent / Destination	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other destinations	Total
USA	22.3	1.0	0.5	0.0	0.6	3.5	1.7	3.8	2.7	7.8	56.2	100.0
GBR	5.6	0.6	3.3	0.0	3.6	3.2	0.6	7.1	0.2	3.9	71.8	100.0
CAN	6.6	2.6	19.5	0.0	3.5	0.9	12.9	5.3	2.6	4.2	41.8	100.0
LIE	3.7	0.4	0.0	0.0	0.9	1.8	0.2	18.7	0.1	3.7	70.5	100.0
DEU	14.8	0.3	1.1	0.0	8.3	1.0	0.0	1.1	0.8	31.9	40.6	100.0
NLD	3.8	0.5	0.2	0.0	0.4	2.2	0.0	0.4	0.4	26.9	65.2	100.0
CHE	5.0	1.0	0.4	0.0	0.7	2.0	0.3	0.5	1.7	6.8	81.6	100.0
JPN	8.7	0.3	0.5	0.0	0.2	0.9	0.0	8.9	1.0	45.0	34.5	100.0
PAN	9.5	0.8	0.6	0.0	1.7	2.0	0.2	1.8	1.4	1.2	80.7	100.0
BRA	14.1	0.5	0.6	0.0	2.3	2.1	2.2	1.6	0.6	18.6	57.4	100.0
Other parents	19.5	0.9	1.6	0.0	2.2	2.1	1.2	2.2	1.7	11.7	56.8	100.0

Row percentages: destination mix of each parent's exports, with the ten parent countries themselves as the destination columns, so the diagonal is the share shipped to the parent's own country. Foreign MNEs with a recorded parent country; top 10 parents by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

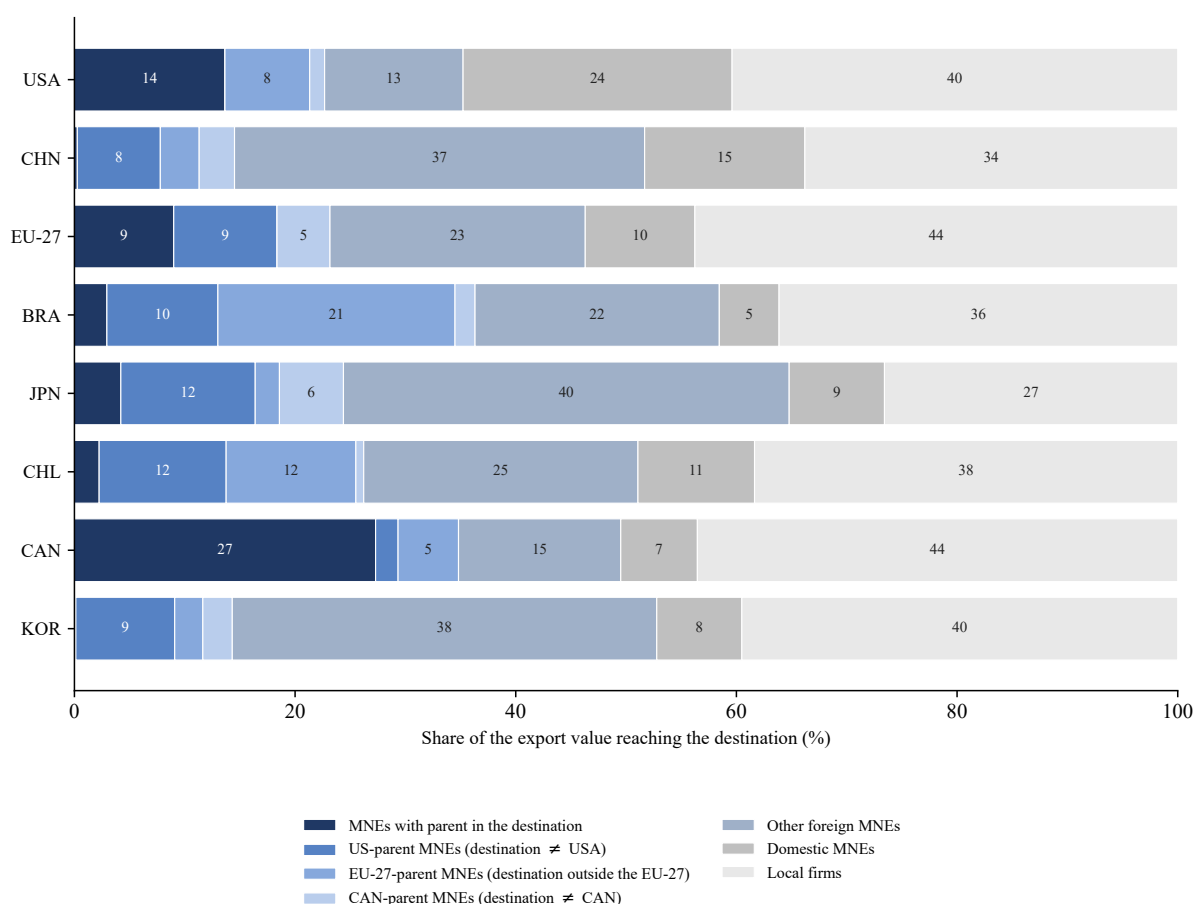


Figure 9: The largest destination markets (EU-27 pooled): share of the value reaching each one moved by MNEs with a parent in the destination, by US-, EU-27- and CAN-parent MNEs (when the destination is not their home), other foreign MNEs, domestic MNEs and local firms.

Note: EU-27 = the 27 member states; the United Kingdom is a separate destination and a separate parent country. The USA row's first segment is the all-goods US-parent share of exports to the United States, column (4) of Table 3.

4.4 Fact 4: A small set of large multinational groups accounts for the bulk of exports

Figure 10 groups the owners of foreign affiliates by the number of affiliates their group has worldwide. Almost half of the owners have a single affiliate, while owners with more than a hundred are 9% of the total and sell 56% of foreign-affiliate export value. The region's foreign-owned export capacity sits inside a small number of large corporate networks.

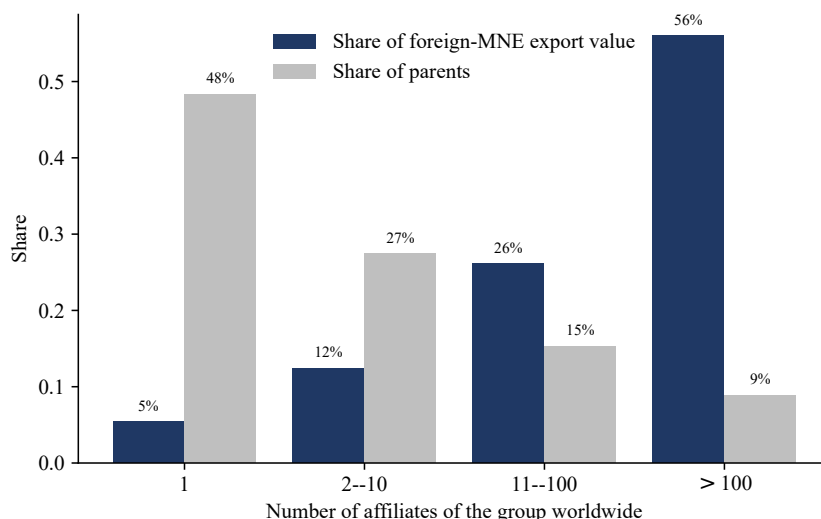


Figure 10: Foreign-MNE export value and number of parents by size of the group's global affiliate network.

4.5 Fact 5: Greater multinational presence is associated with higher trade volumes

Table 9 relates the exports of an origin, destination, product and year cell to the multinational presence in it, on the intensive margin (how many affiliates sell the product there) and on the extensive one (whether any does). Both associations are large and survive an increasingly demanding set of fixed effects, and both hold when the dependent variable is the exports of the *non-multinational* firms in the same cell (Panel B).

Table 10 splits the intensive margin by owner. The coefficient is about three times larger for foreign affiliates than for domestic ones, 1.21 against 0.44, and the same gap appears for the exports of local firms in the cell, 1.13 against 0.47. On the extensive margin the two are almost identical, 1.50 against 1.47 (Table 11), so what separates foreign affiliates is how much presence they bring rather than whether they are there at all. Columns (5)–(6) split foreign affiliates further by whether the group reaches the destination through its own headquarters; they rest on far fewer cells and should be read as suggestive.

Table 9: Multinational presence and trade volume: intensive and extensive margins

	Intensive: ln(# MNE firms)			Extensive: any MNE present		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A: all exports (ln)</i>						
ln(# MNE firms)	2.0823*** (0.0363)	2.0685*** (0.0368)	0.9414*** (0.0206)			
Any MNE present				1.4622*** (0.0394)	1.4591*** (0.0388)	0.6818*** (0.0165)
<i>Panel B: non-MNE exports (ln)</i>						
ln(# MNE firms)	1.3536*** (0.0318)	1.3406*** (0.0319)	0.2389*** (0.0178)			
Any MNE present				1.4168*** (0.0371)	1.3996*** (0.0367)	0.1311*** (0.0102)
Origin FE	✓			✓		
Destination FE	✓			✓		
Product FE	✓	✓		✓	✓	
Year FE	✓			✓		
Origin × year FE		✓			✓	
Destination × year FE		✓			✓	
Origin × dest. × product FE			✓			✓
Origin × dest. × year FE			✓			✓
Observations (all exports)	1,008,472	1,008,185	883,545	1,829,852	1,829,626	1,633,319
Observations (non-MNE exports)	475,856	475,514	429,882	1,297,322	1,297,111	1,137,550

Origin-destination-product-year cells; dep. var. ln exports. Intensive margin: ln number of MNE firms (cells with ≥ 1 MNE); extensive margin: indicator for any MNE present. SE clustered at origin-destination. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 10: Intensive margin: ln(# MNE firms) decomposed into foreign / domestic and foreign through HQ / not through HQ

	All MNEs		Foreign / domestic		Foreign through HQ / not through HQ / domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
ln(# MNE firms)	2.0685*** (0.0368)	0.9414*** (0.0206)				
ln(# foreign MNEs)			1.2129*** (0.0285)	0.4506*** (0.0124)		
ln(# foreign MNEs present through HQ)					0.3970*** (0.0856)	0.1448*** (0.0370)
ln(# foreign MNEs not present through HQ)					0.9465*** (0.0536)	0.3133*** (0.0382)
ln(# domestic MNEs)			0.4382*** (0.0259)	0.2214*** (0.0151)	0.3770*** (0.0514)	0.1555*** (0.0315)
<i>Panel B: non-MNE exports (ln)</i>						
ln(# MNE firms)	1.3406*** (0.0319)	0.2389*** (0.0178)				
ln(# foreign MNEs)			1.1331*** (0.0375)	0.2413*** (0.0217)		
ln(# foreign MNEs present through HQ)					0.2934*** (0.0882)	0.1945*** (0.0527)
ln(# foreign MNEs not present through HQ)					1.0330*** (0.0607)	0.1951*** (0.0648)
ln(# domestic MNEs)			0.4707*** (0.0341)	0.1250*** (0.0218)	0.4142*** (0.0633)	0.0946* (0.0552)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (all exports)	1,008,185	883,545	113,422	98,223	9,197	7,524
Observations (non-MNE exports)	475,514	429,882	93,010	81,563	8,780	7,255

Origin-destination-product-year cells; dep. var. ln exports. Intensive margin: plain logs of the counts, so each column uses the cells where every count entering it is positive (no $\ln(1+x)$). (1)–(2) all MNEs; (3)–(4) foreign vs domestic; (5)–(6) foreign split by whether the group is present at the destination through its headquarters (the parent's country). SE clustered at origin-destination. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 11: Extensive margin: presence indicators decomposed into foreign / domestic and foreign through HQ / not through HQ

	All MNEs		Foreign / domestic		Foreign through HQ / not through HQ / domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
Any MNE present	1.4591*** (0.0388)	0.6818*** (0.0165)				
Any foreign MNE			1.4990*** (0.0339)	0.6986*** (0.0158)		
Any foreign MNE present through HQ					1.3873*** (0.0951)	0.5909*** (0.0431)
Any foreign MNE not present through HQ					1.5093*** (0.0319)	0.6976*** (0.0146)
Any domestic MNE			1.4670*** (0.0416)	0.4917*** (0.0165)	1.4334*** (0.0388)	0.4909*** (0.0166)
<i>Panel B: non-MNE exports (ln)</i>						
Any MNE present	1.3996*** (0.0367)	0.1311*** (0.0102)				
Any foreign MNE			1.2983*** (0.0346)	0.1356*** (0.0106)		
Any foreign MNE present through HQ					1.0183*** (0.0900)	0.1265*** (0.0194)
Any foreign MNE not present through HQ					1.2782*** (0.0326)	0.1386*** (0.0107)
Any domestic MNE			1.1316*** (0.0428)	0.1370*** (0.0138)	1.0959*** (0.0403)	0.1367*** (0.0138)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (all exports)	1,829,626	1,633,319	1,829,626	1,633,319	1,829,626	1,633,319
Observations (non-MNE exports)	1,297,111	1,137,550	1,297,111	1,137,550	1,297,111	1,137,550

One observation = origin *imes* destination *imes* HS6 product *imes* year; dep. var. ln exports of the product from the origin to the destination in the year. Extensive margin: indicators for at least one MNE of each type exporting the product to the destination; all observations. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

4.6 Fact 6: Distance is a weaker barrier to trade for multinational corporations

Table 12 estimates the distance elasticity of firm-level exports and how it changes for multinational affiliates. The baseline elasticity is -0.16 and roughly a third of it goes away for multinationals. Table 13 splits that attenuation: 0.049 for foreign affiliates against 0.032 for domestic ones, and 0.064 for foreign affiliates whose group reaches the destination through its headquarters, which is to say for affiliates selling to their own parent's country.

Table 12: Distance and firm exports: multinationals split by presence at the destination

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1639*** (0.0454)	-0.1848*** (0.0460)	-0.1632*** (0.0452)	-0.1838*** (0.0459)
ln distance × MNE	0.0461*** (0.0038)	0.0458*** (0.0038)		
ln distance × MNE, present			0.0689*** (0.0070)	0.0682*** (0.0070)
ln distance × MNE, not present			0.0393*** (0.0032)	0.0391*** (0.0032)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. multinationals (foreign or domestic) split by presence at the destination (the group has an affiliate there, or the destination is the parent's country). SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

Table 13: Distance and firm exports: foreign / domestic, then foreign through HQ / not through HQ

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1645*** (0.0454)	-0.1853*** (0.0460)	-0.1644*** (0.0454)	-0.1851*** (0.0460)
ln distance × foreign MNE	0.0487*** (0.0041)	0.0483*** (0.0040)		
ln distance × foreign MNE, present through HQ			0.0642*** (0.0159)	0.0638*** (0.0161)
ln distance × foreign MNE, not present through HQ			0.0477*** (0.0035)	0.0473*** (0.0035)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0318*** (0.0042)	0.0317*** (0.0042)	0.0317*** (0.0042)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. (1)–(2) foreign vs domestic multinationals; (3)–(4) foreign MNEs split by whether the group is present at the destination through its headquarters (the parent's country) or not. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

5 Products and Sectors

The facts above average over a very uneven product range. Figure 11 shows the twenty largest HS6 lines with the foreign, domestic and local shares of each. Crude petroleum is three quarters domestically owned, copper concentrates are four fifths foreign owned, gold sits in between. Figure 12 splits the foreign bar by owner country and gives a readable map of who owns what: British groups in copper and coal, Canadian groups in gold, US groups in soybeans, maize and apparel, German groups in trucks, Swiss groups in soybean cake. Table 14 aggregates to HS sections with the leading owner of each.

Tables 15 to 18 give the same breakdowns within each sector: agriculture by SITC Rev. 3 division, manufacturing by HS section and by BEC end use, mining and fuels by BEC end use.

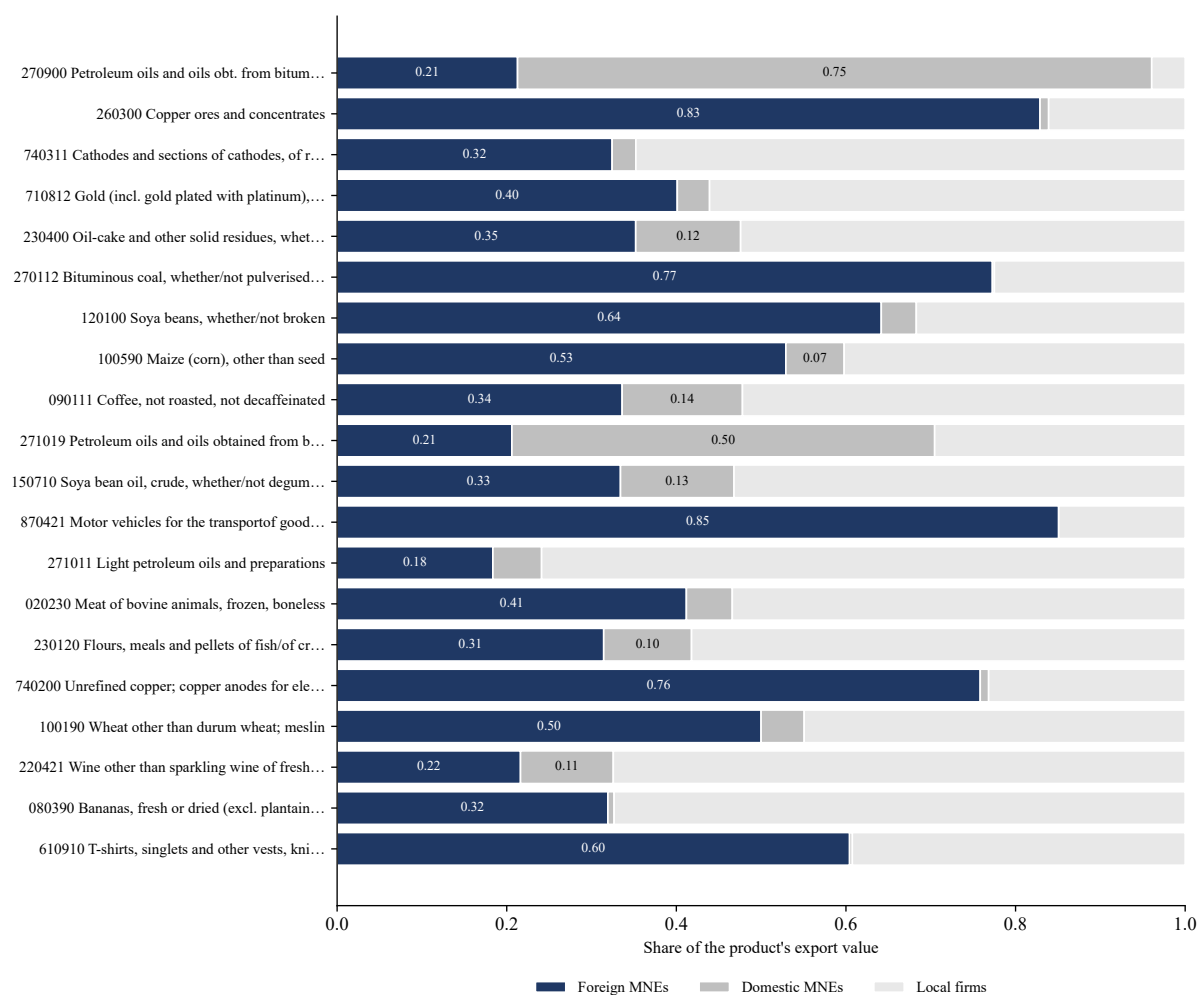


Figure 11: Top-20 HS6 products by export value: foreign MNE, domestic MNE and local shares.

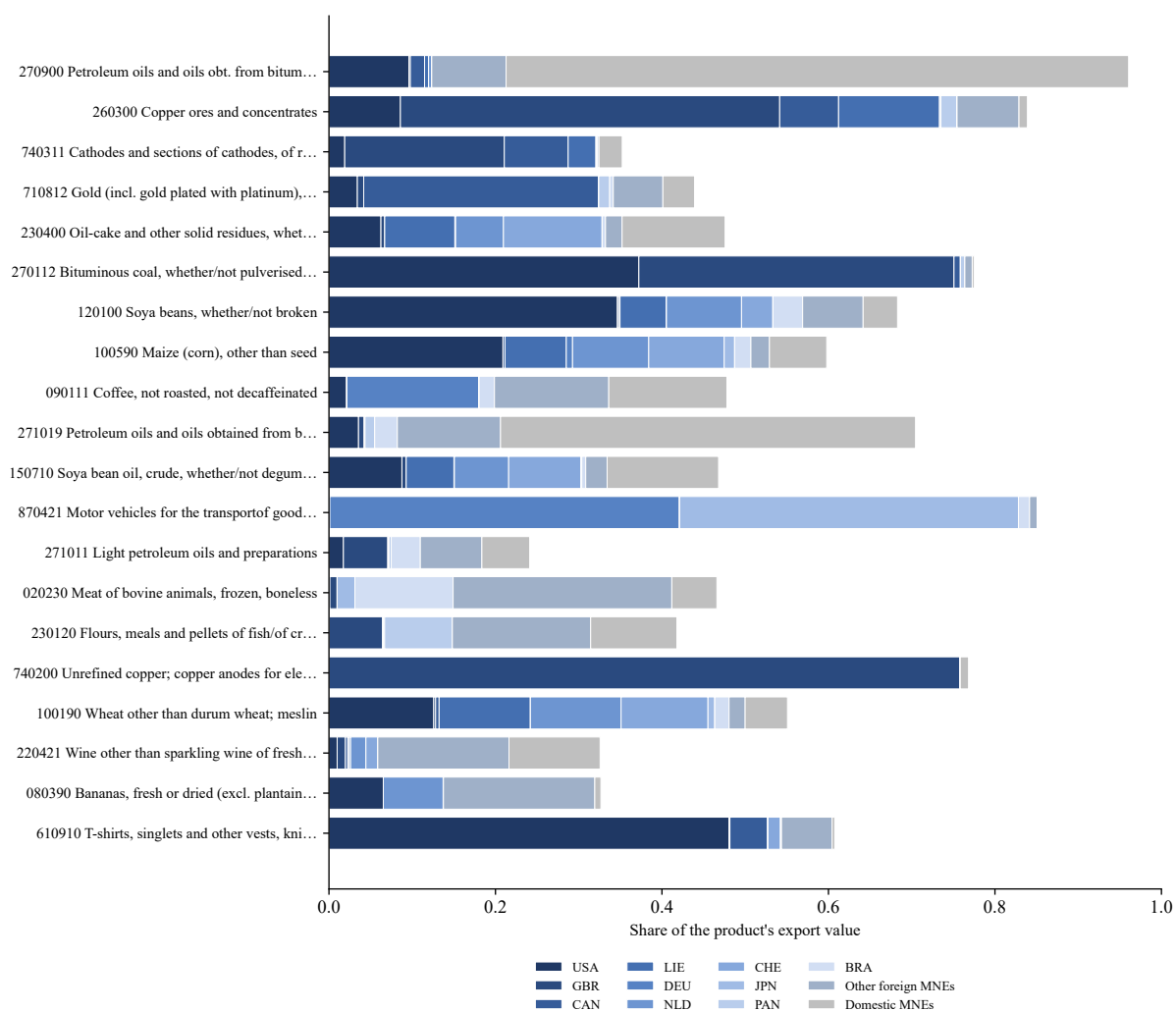


Figure 12: Top-20 HS6 products: foreign bar split by the parent's country (remainder = local firms).

Table 14: HS sections by export value: MNE shares and leading parent

HS section	\$bn/yr	For.	Dom.	Local	Lead parent
V Mineral products	57.1	51	29	19	GBR (0.42)
II Vegetable products	33.5	42	7	51	USA (0.33)
IV Prepared foodstuffs, beverages, tobacco	25.9	37	10	53	USA (0.21)
XV Base metals	19.6	41	3	56	GBR (0.44)
XIV Precious metals and stones	15.3	41	4	56	CAN (0.65)
I Live animals; animal products	12.5	34	14	51	SAU (0.19)
VI Chemicals	11.6	49	14	37	USA (0.25)
XVII Transport equipment	9.7	78	4	18	DEU (0.30)
XI Textiles and apparel	6.9	49	3	48	USA (0.62)
III Fats and oils	6.9	31	10	59	CHE (0.26)
XVI Machinery and electrical equipment	6.2	63	4	33	USA (0.41)
VII Plastics and rubber	5.8	51	8	42	USA (0.34)
X Pulp, paper	3.4	54	28	18	PAN (0.54)
XVIII Optical, precision instruments	3.4	72	8	19	USA (0.66)
IX Wood and articles	1.9	27	34	39	BHS (0.24)
VIII Hides, skins, leather	1.6	40	12	49	BGR (0.19)
XIII Stone, ceramics, glass	0.9	49	4	46	USA (0.32)
XX Miscellaneous manufactures	0.9	53	7	40	USA (0.48)
Unclassified	0.5	49	29	23	USA (0.39)
XII Footwear, headgear	0.5	48	1	50	USA (0.66)
XXI Works of art	0.1	58	1	41	BGR (0.94)
XIX Arms	0.0	1	0	99	CHE (0.82)
Special / unclassified (98-99)	0.0	13	1	87	USA (0.57)
All sections	224.5	46	14	40	

HS sections ranked by export value. For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the section's export value, percent; Lead parent = largest parent country among the section's foreign MNEs and its share of the section's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 15: Agriculture by SITC Rev. 3 division: total and MNE exports, shares, leading parents

SITC Rev. 3 division	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
08 Animal feed	13.8	6.6	35.7	11.5	CHE 0.28, LIE 0.19, USA 0.18
05 Vegetables & fruit	10.6	4.0	32.6	5.8	ESP 0.14, USA 0.13, PAN 0.11
04 Cereals	10.3	5.9	48.6	8.2	USA 0.33, CHE 0.14, NLD 0.14
22 Oil seeds	6.7	4.4	61.6	4.1	USA 0.54, NLD 0.14, LIE 0.08
07 Coffee, tea, cocoa, spices	5.1	2.3	30.4	15.6	DEU 0.41, ITA 0.14, SGP 0.12
42 Fixed vegetable oils	6.0	2.5	30.8	10.2	CHE 0.27, NLD 0.24, USA 0.23
01 Meat	5.2	2.4	34.8	13.1	SAU 0.46, BRA 0.26, JPN 0.08
03 Fish & seafood	3.4	1.8	41.9	9.5	ESP 0.32, LUX 0.11, ITA 0.10
29 Crude animal/veg. materials	2.2	0.8	27.3	9.0	DEU 0.10, ARG 0.09, BGR 0.08
11 Beverages	2.2	0.8	28.6	9.2	LUX 0.17, PAN 0.15, GBR 0.12
02 Dairy & eggs	1.9	1.0	28.0	20.3	NLD 0.20, CHE 0.19, ROU 0.19
09 Misc. edible products	1.7	1.2	70.9	2.3	USA 0.46, CHE 0.17, FRA 0.09
06 Sugar & honey	1.4	0.5	19.3	14.7	USA 0.50, KOR 0.18, LIE 0.10
12 Tobacco	1.2	0.7	54.8	0.2	USA 0.36, HKG 0.25, GBR 0.12
41 Animal oils & fats	0.5	0.2	23.0	11.0	BMU 0.31, PAN 0.23, GBR 0.16
Other	0.6	0.1	21.6	1.9	CHE 0.21, USA 0.20, ROU 0.10
All	72.9	35.0	38.5	9.4	

Agriculture (HS 01–24), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 16: Manufacturing by HS section: total and MNE exports, shares, leading parents

HS section	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
VI Chemicals	11.6	7.4	49.3	14.0	USA 0.25, DEU 0.13, GBR 0.12
VII Plastics and rubber	5.8	3.4	50.5	7.7	USA 0.34, MEX 0.17, JPN 0.06
VIII Hides, skins, leather	1.6	0.8	39.7	11.6	BGR 0.19, URY 0.17, SGP 0.16
IX Wood and articles	1.9	1.1	27.4	33.9	BHS 0.24, JPN 0.12, VGB 0.11
X Pulp, paper	3.4	2.7	54.2	28.2	PAN 0.54, USA 0.19, CHL 0.08
XI Textiles and apparel	6.9	3.6	49.1	2.9	USA 0.62, CAN 0.04, FRA 0.03
XII Footwear, headgear	0.5	0.3	48.2	1.3	USA 0.66, BRA 0.13, JPN 0.08
XIII Stone, ceramics, glass	0.9	0.5	49.5	4.5	USA 0.32, PAN 0.20, CYM 0.11
XV Base metals	19.6	8.5	41.1	3.1	GBR 0.44, CAN 0.12, ZAF 0.08
XVI Machinery and electrical equipment	6.2	4.2	63.2	3.9	USA 0.41, IRL 0.10, JPN 0.08
XVII Transport equipment	9.7	7.9	77.6	4.2	DEU 0.30, JPN 0.28, NLD 0.12
XVIII Optical, precision instruments	3.4	2.8	72.3	8.5	USA 0.66, DEU 0.12, GBR 0.08
XX Miscellaneous manufactures	0.9	0.6	52.9	6.9	USA 0.48, SWE 0.10, DEU 0.10
Other	0.1	0.0	38.0	0.5	BGR 0.92, USA 0.03, CHE 0.01
All	72.7	43.9	51.6	8.0	

Manufacturing (HS 28–97 excl. 71), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 17: Manufacturing by BEC end use: total and MNE exports, shares, leading parents

BEC end use	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Intermediate	47.2	26.6	46.9	9.1	GBR 0.20, USA 0.20, PAN 0.07
Consumption	15.2	10.0	60.6	4.9	USA 0.43, NLD 0.09, DEU 0.08
Capital	9.3	6.7	64.8	5.6	DEU 0.31, JPN 0.28, USA 0.18
All	71.7	43.3	51.7	7.8	

Manufacturing (HS 28–97 excl. 71), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 18: Mining and fuels by BEC end use: total and MNE exports, shares, leading parents

BEC end use	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Intermediate	65.4	48.6	51.6	23.7	GBR 0.37, USA 0.21, CAN 0.16
Consumption	0.5	0.3	61.7	0.4	USA 0.22, FRA 0.22, BEL 0.21
Capital	0.0	0.0	21.4	0.0	FRA 0.82, USA 0.08, GBR 0.03
All	65.9	48.9	51.6	23.6	

Mining & fuels (HS 25–27, 71), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

6 Theoretical Framework

To be written. The framework is the companion theory paper: an oligopoly model with multinational ownership in which the object of interest is the ownership-weighted Herfindahl rather than the country-level ownership share. The section will state the environment, the objects the facts discipline, and how the model's ownership share maps to the measured one.

7 Concluding Remarks

To be written.

References

To be completed with the literature section. The sources of the classifications used above are:

- Fontagné, L., Guimbard, H. and Orefice, G., 2022. Tariff-Based Product-Level Trade Elasticities. *Journal of International Economics*, 137.
- Hausmann, R. and Hidalgo, C., 2011. The Network Structure of Economic Output. *Journal of Economic Growth*, 16(4).
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- United Nations Statistics Division, 2022. *Classification by Broad Economic Categories*, Rev. 4. New York.

A Appendix: Additional Exhibits

This appendix collects the exhibits that document or extend the figures and tables of the main text, and the numbers behind them, in the order of the facts.

A.1 Fact 1

Table A.1: Numbers behind Figure 1: foreign, domestic and total MNE shares by origin

Origin	Foreign	Domestic	MNE total	Value (\$bn/yr)
COL	0.353	0.391	0.744	44.0
DOM	0.684	0.013	0.697	8.6
CRI	0.597	0.036	0.633	10.5
URY	0.528	0.100	0.628	7.0
PRY	0.611	0.011	0.622	7.3
CHL	0.479	0.136	0.615	34.2
ARG	0.489	0.075	0.564	66.4
SLV	0.474	0.003	0.477	5.0
PER	0.442	0.023	0.466	41.4
All	0.463	0.141	0.605	224.5

Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

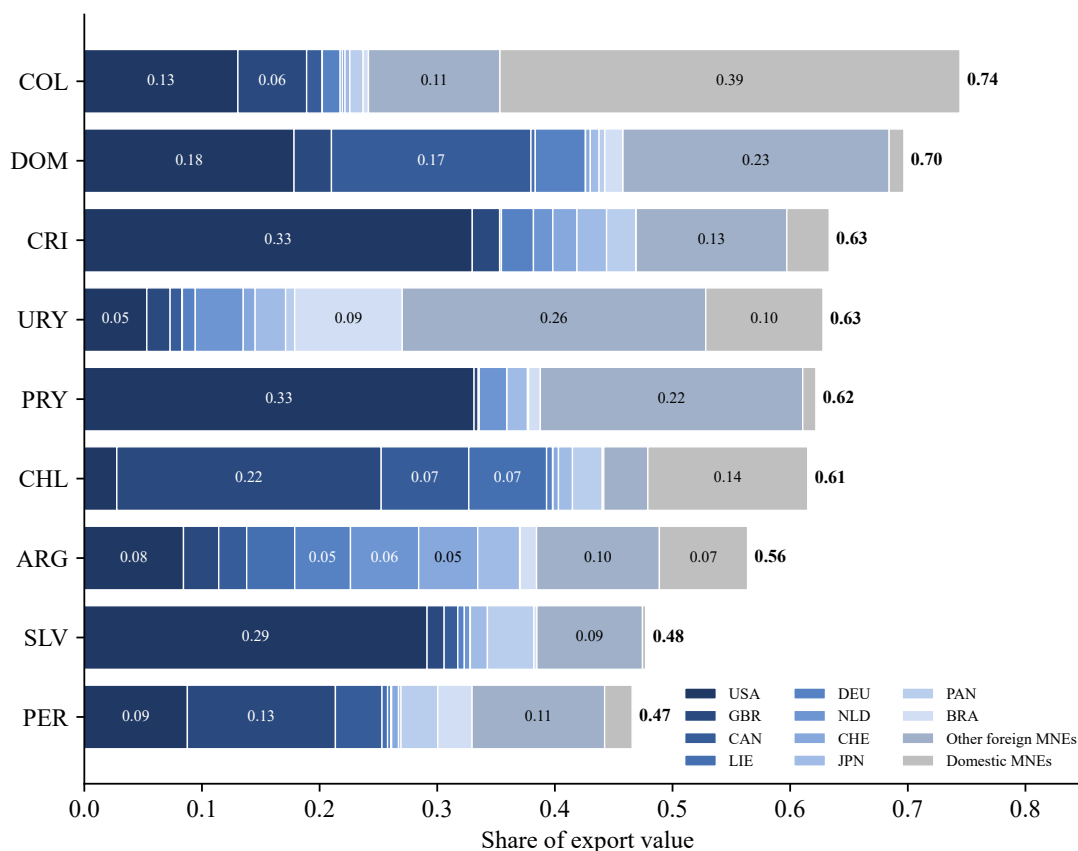
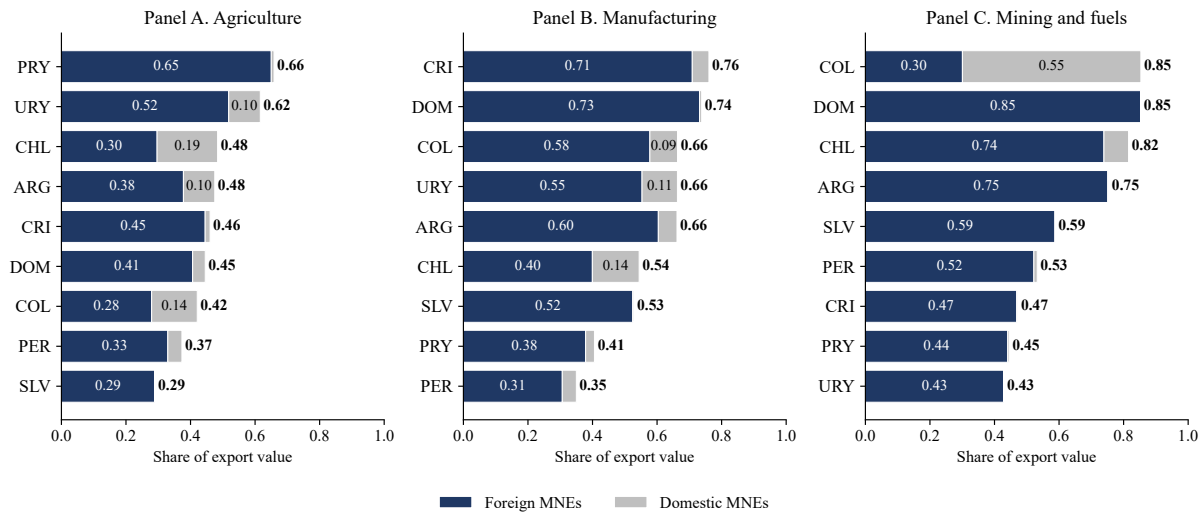


Figure A.1: MNE share of export value by origin, foreign bar split by the parent's country (top 10, other foreign, domestic).

Table A.2: Numbers behind the previous figure

Origin	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Dom.	MNE total	Value (\$bn/yr)
COL	0.131	0.058	0.013	0.000	0.015	0.002	0.002	0.004	0.011	0.005	0.112	0.391	0.744	44.0
DOM	0.178	0.032	0.170	0.004	0.043	0.000	0.004	0.007	0.005	0.015	0.226	0.013	0.697	8.6
CRI	0.330	0.023	0.002	0.000	0.027	0.016	0.020	0.025	0.025	0.000	0.128	0.036	0.633	10.5
URY	0.053	0.020	0.010	0.000	0.011	0.041	0.010	0.026	0.008	0.091	0.258	0.100	0.628	7.0
PRY	0.331	0.004	0.000	0.000	0.000	0.023	0.000	0.017	0.001	0.010	0.223	0.011	0.622	7.3
CHL	0.028	0.225	0.074	0.066	0.005	0.000	0.005	0.012	0.025	0.001	0.037	0.136	0.615	34.2
ARG	0.084	0.030	0.024	0.041	0.047	0.058	0.050	0.036	0.001	0.014	0.104	0.075	0.564	66.4
SLV	0.291	0.015	0.012	0.000	0.006	0.005	0.000	0.014	0.040	0.003	0.090	0.003	0.477	5.0
PER	0.088	0.126	0.040	0.005	0.003	0.001	0.006	0.002	0.031	0.029	0.113	0.023	0.466	41.4

Figure 1 split by parent country. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country. Dollar values are annual averages: each origin’s pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

**Figure A.2:** MNE share of export value by origin within each sector: foreign and domestic multinationals (Panel A agriculture, Panel B manufacturing, Panel C mining and fuels).**Table A.3:** The four sectors: total and MNE exports, foreign and domestic shares, leading parents

Sector	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Agriculture	78.9	37.8	38.1	9.5	USA 0.24, CHE 0.11, NLD 0.10
Mining & fuels	72.4	52.4	49.2	24.5	GBR 0.36, USA 0.21, CAN 0.16
Manufacturing	72.7	43.9	51.6	8.0	USA 0.25, GBR 0.14, DEU 0.09
Rest	0.5	0.4	47.9	28.0	USA 0.39, GBR 0.12, ESP 0.11
All	224.5	134.5	46.3	14.1	

Agriculture = HS 01–24; Mining & fuels = HS 25–27 and 71 (precious metals and stones); Manufacturing = HS 28–97 (excl. 71); Rest = HS 98–99, chapter 00 and unclassifiable codes. Services are not observable in customs merchandise data. Pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row’s export value; leading parents = the three largest parent countries among the row’s foreign MNEs with their share of the row’s foreign-MNE value. Dollar values are annual averages: each origin’s pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

A.2 Fact 2

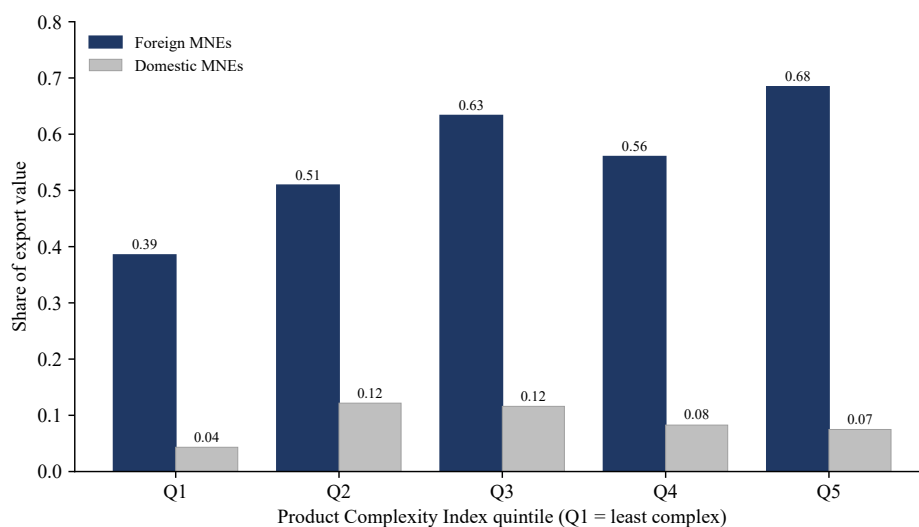


Figure A.3: Manufacturing: MNE share of export value by PCI quintile (quintiles over manufacturing HS6 products).

A.3 Fact 3

Table A.4: Parent region × destination region: destination mix of each group (row %)

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	40.4	15.5	14.7	27.2	1.2	0.7	0.3	100.0
North America	31.3	23.6	20.0	23.1	1.7	0.1	0.1	100.0
Europe	28.0	11.7	17.8	38.4	2.9	1.0	0.1	100.0
Asia	51.0	18.7	11.7	17.0	1.1	0.2	0.3	100.0
Africa	10.9	6.6	26.9	53.2	2.2	0.0	0.2	100.0
Oceania	20.1	5.7	8.4	65.5	0.1	0.2	0.0	100.0
Rest of World	32.6	25.0	20.9	14.3	0.0	7.1	0.0	100.0
Domestic MNEs	25.9	32.9	11.1	27.0	2.3	0.5	0.3	100.0
Local firms	27.1	21.6	21.7	25.7	2.9	0.6	0.4	100.0

Row percentages: destination mix of each row group. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

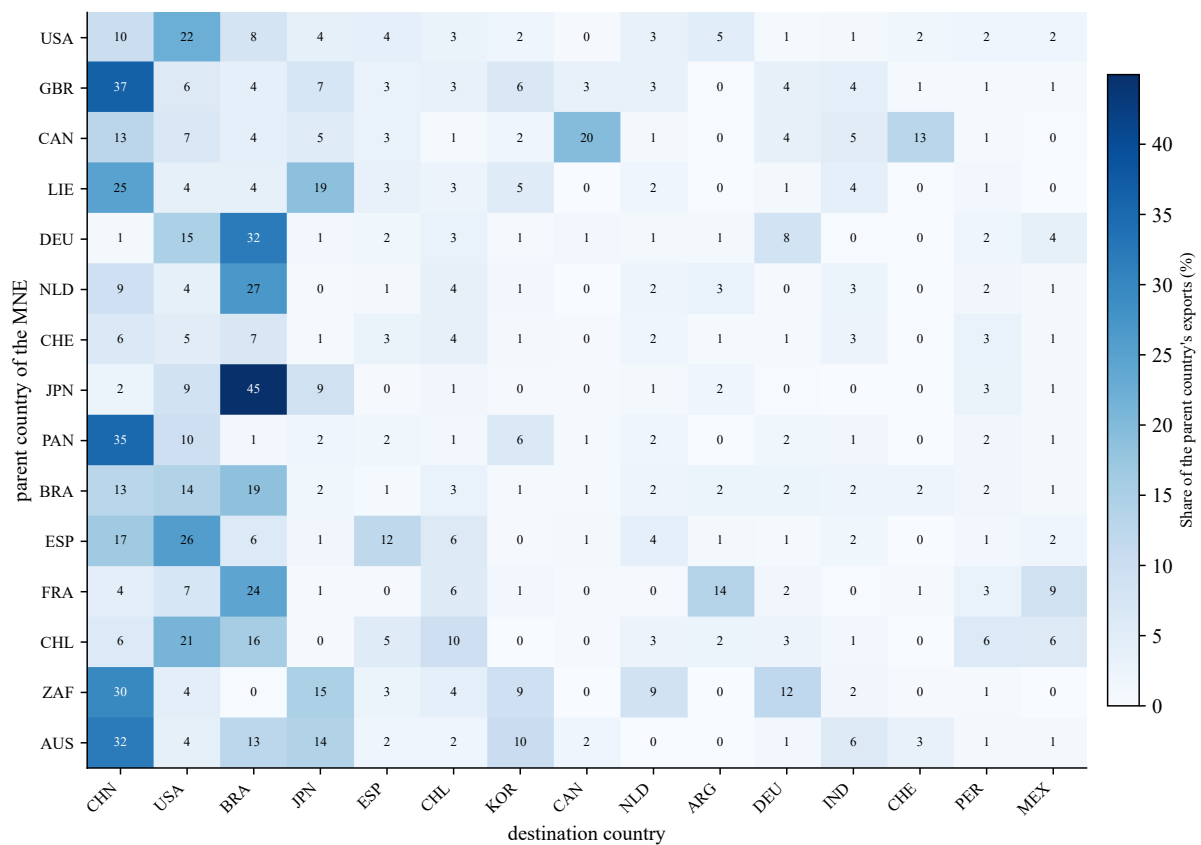


Figure A.4: Top-15 parents × top-15 destinations: % of the parent's export value going to each destination.

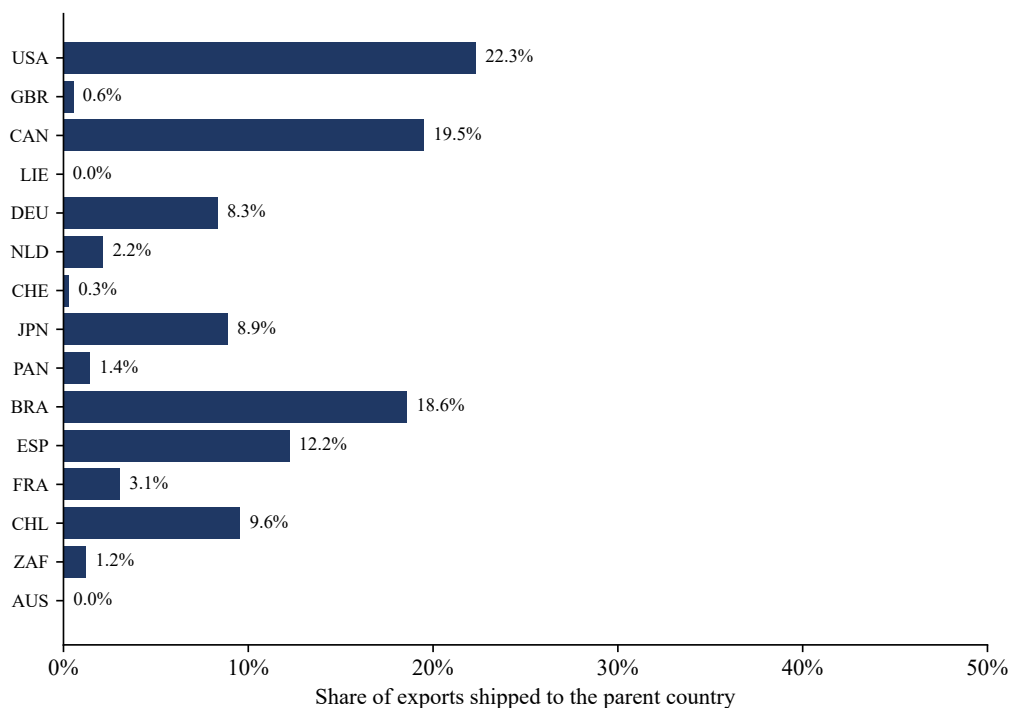


Figure A.5: Share of each parent's export value shipped to the parent's own country, parents as recorded (the diagonal of Table 8).

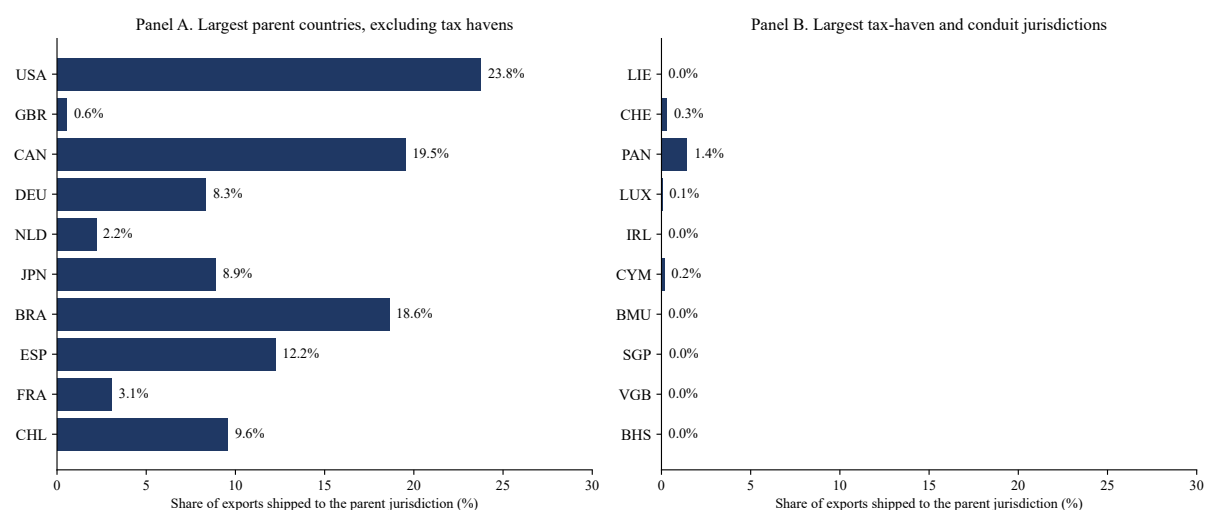


Figure A.6: Share of each parent’s export value shipped to the parent’s own jurisdiction: Panel A, the ten largest parent countries that are not tax havens (dependencies folded into their sovereign); Panel B, the ten largest tax-haven and conduit jurisdictions.

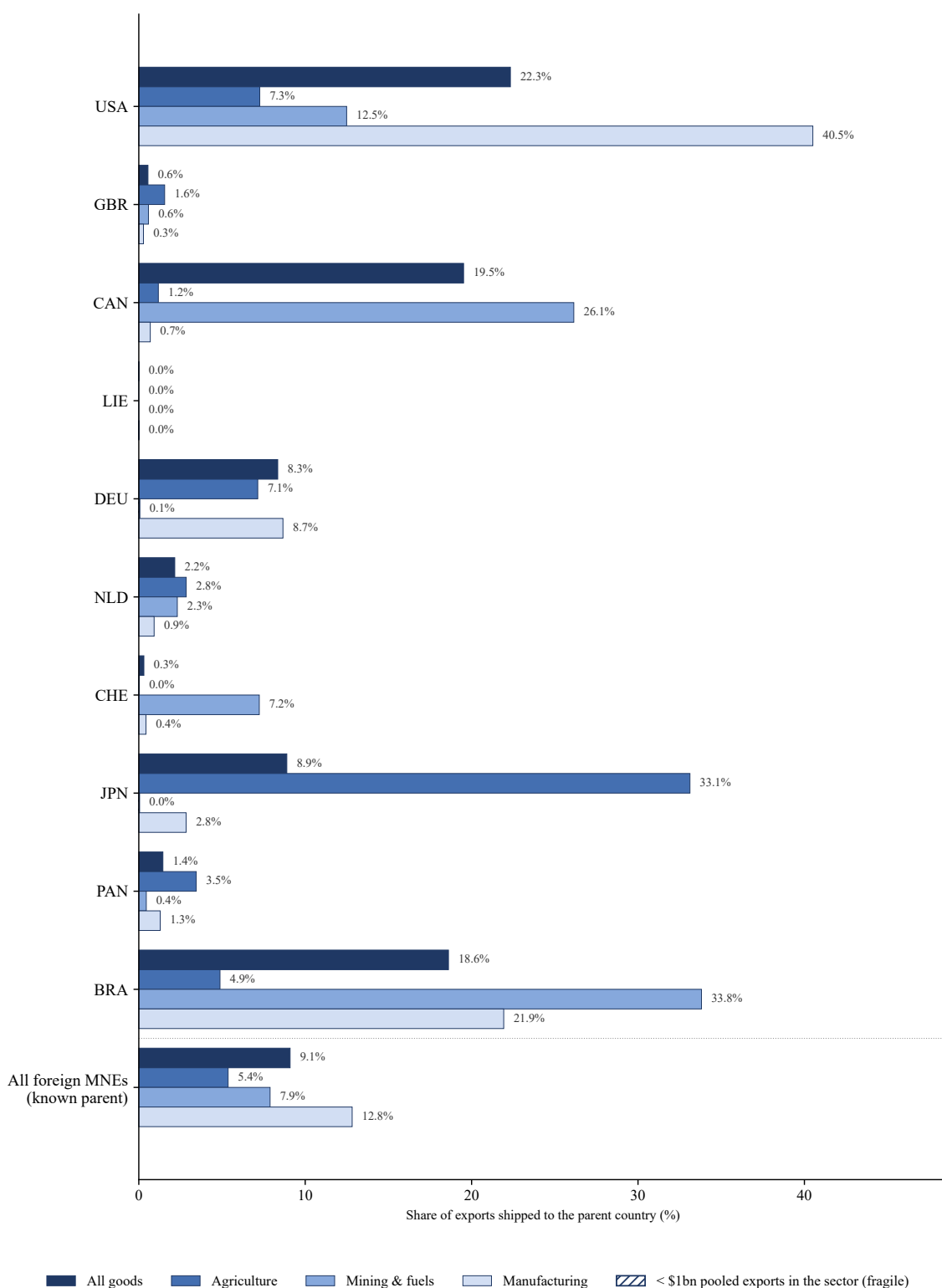


Figure A.7: Share of each parent's export value shipped to the parent's own country: all goods and the three sectors, top-10 parents.

Note: Hatched bars: the parent's pooled exports in the sector are below \$1bn, so the share is fragile.

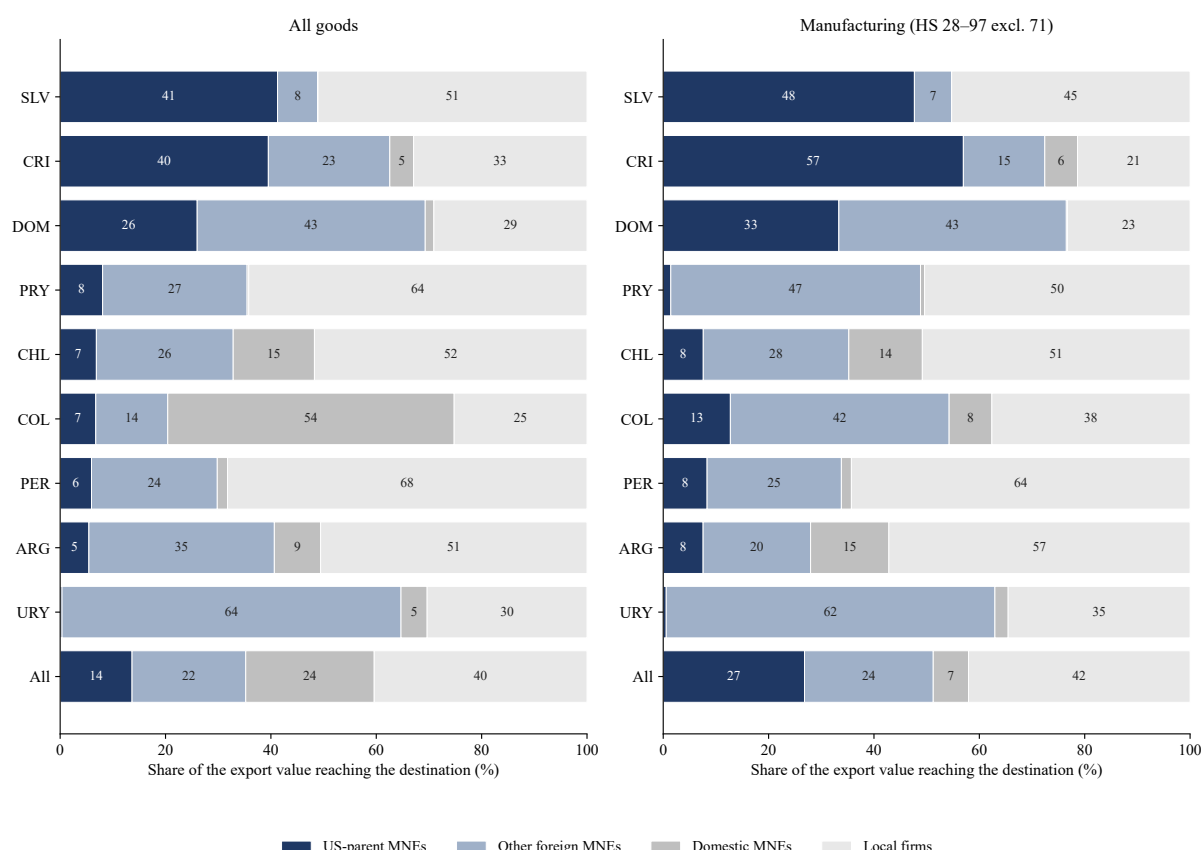


Figure A.8: Exports to the United States by origin: share moved by US-parent MNEs, other foreign MNEs, domestic MNEs and local firms: all goods (left) and manufacturing (right).

Note: Denominator: the origin's exports to the United States. The US-parent bars of the left panel are column (4) of Table 3.

Table A.5: Exports to the United States by sector: who carries them (%)

Sector	Exports to the USA (\$bn/yr)	US-parent MNEs (%)	Other foreign MNEs (%)	Domestic MNEs (%)	Local firms (%)
All goods	40.0	13.6	21.6	24.4	40.4
Agriculture	10.7	4.8	28.0	9.2	58.0
Mining & fuels	14.6	6.6	14.8	52.0	26.6
Manufacturing	14.5	26.8	24.4	6.7	42.1
Rest	0.1	40.0	38.4	4.5	17.1

Exports from the nine origins to the United States by sector, and the share of that value moved by each type of exporter. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table A.6: What US-parent MNEs ship to the United States: top-15 HS6 lines, the US-parent share of everything the nine origins export of the line to the USA, and the main origins

HS6	Description	US-parent exports to the USA (\$bn/yr)	% of all US-parent exports to the USA	% of US-parent exports of the line going to the USA	All LAC exports of the line to the USA (\$bn/yr)	US-parent share of LAC exports of the line to the USA (%)	Main origins of the US-parent shipments (% of the line)
610910	T-shirts, singlets & other vests, knitted/cro...	0.49	11.1	84	1.01	53	SLV 85%, DOM 10%, PER 5%
901839	Catheters, cannulae & the like	0.52	9.6	73	0.63	89	CRI 92%, DOM 8%, COL 0%
270900	Petroleum oils & oils obt. from bituminous mi...	0.41	9.1	24	7.84	5	COL 100%, ARG 0%
901890	Instruments & appliances used in medical/surg...	0.39	6.8	69	0.94	47	CRI 71%, DOM 29%, COL 0%
270112	Bituminous coal, whether/not pulverised but n...	0.19	4.3	9	0.39	48	COL 100%
854231	Electronic integrated circuits, processors & ...	0.21	3.9	28	0.22	95	CRI 100%, COL 0%, ARG 0%
401110	New pneumatic tyres, of rubber, of a kind use...	0.13	3.4	93	0.24	63	CHL 100%, DOM 0%, COL 0%
261390	Molybdenum ores & concentrates, other than ro...	0.15	2.8	99	0.18	79	PER 100%
611595	Other n.e.s. in 61.15, knitted/crocheted, of ...	0.10	2.4	100	0.11	97	SLV 100%, DOM 0%
610711	Men's/boys' underpants & briefs, knitted/croc...	0.09	2.2	87	0.10	87	SLV 100%, DOM 0%, CRI 0%
621210	Brassieres & parts thereof, whether/not knit...	0.09	1.8	98	0.12	78	SLV 74%, DOM 26%, CRI 0%
080430	Pineapples, fresh/dried	0.08	1.6	66	0.42	20	CRI 100%
240210	Cigars, cheroots & cigarillos containing toba...	0.09	1.3	100	0.57	16	DOM 100%
740311	Cathodes & sections of cathodes, of refined c...	0.06	1.2	24	1.56	3	PER 64%, CHL 36%
640510	Footwear with uppers of leather/composition l...	0.08	1.2	70	0.11	71	DOM 100%
All lines		5.0	100.0	22	40.0	14	

Top 15 HS6 lines by the value US-parent multinationals export from the nine origins to the United States. Column 5 = of what US-parent MNEs export of the line from LAC (all destinations), the share shipped to the USA; column 7 = of what all firms in the nine origins export of the line to the USA, the share moved by US-parent MNEs; last column = the three origins with the largest US-parent shipments of the line to the USA and their share of those shipments. Dollar columns: recorded US parents only. Foreign MNEs whose parent country is not recorded (7.5% of foreign-MNE export value) are allocated to parent countries in proportion to the recorded parents of the same group, so every share by parent country refers to all foreign MNEs and adds up with the domestic and local shares. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

A.4 Fact 4

Table A.7: Parents and foreign-MNE export value by global affiliate-network size

Global affiliate count	Parents	Share of parents (%)	Share of foreign-MNE export value (%)
1	5,982	48.4	5.4
2–10	3,397	27.5	12.4
11–100	1,893	15.3	26.2
>100	1,099	8.9	56.0
All parents with a network record	12,371	100	100

Foreign-MNE parents (parent name key) exporting from the nine origins; 98% of parents and 97% of foreign-MNE export value have a matched global-network record (Orbis $\cup$ D&B roster, src/16). Network size = number of distinct worldwide affiliates.

A.5 Fact 5

Table A.8: MNE export value by origin: domestic share, and foreign-MNE value by the group's presence at the destination

Origin	MNE value (\$bn/yr)	of which domestic (%)	Foreign-MNE value (\$bn/yr)	% of foreign-MNE value: group present at destination	
				through headquarters	not through HQ (of which via another affiliate)
ARG	37.5	13.3	32.5	6.5	93.5 (41.8)
CHL	21.0	22.1	16.4	4.3	95.7 (56.7)
COL	32.8	52.5	15.5	6.5	93.5 (34.2)
CRI	6.6	5.7	6.3	25.5	74.5 (33.8)
DOM	6.0	1.8	5.9	29.0	71.0 (21.6)
PER	19.3	5.0	18.3	3.9	96.1 (28.8)
PRY	4.5	1.8	4.4	2.1	97.9 (65.5)
SLV	2.4	0.6	2.4	40.7	59.3 (28.1)
URY	4.4	15.9	3.7	5.2	94.8 (37.7)
All	134.5	23.4	105.4	8.4	91.6 (40.5)

'Through headquarters': the affiliate ships to its parent's country. 'Not through HQ': every other destination; in parentheses the part where the group has another affiliate in the destination (Orbis/D&B roster). Pooled 2006–2022. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table A.9: Fact 5 by sector, intensive margin: number of foreign and of domestic MNEs exporting the product to the destination, all goods and the three sectors

Sample	All goods	Agriculture	Mining & fuels	Manufacturing
<i>Panel A. Dependent variable: ln exports of the product from the origin to the destination in the year; all firms</i>				
Fixed effects: origin × year, destination × year, product				
Number of foreign MNEs exporting the product to the destination (ln)	1.2129*** (0.0285)	1.2954*** (0.0275)	1.1301*** (0.1799)	1.1644*** (0.0358)
Number of domestic MNEs exporting the product to the destination (ln)	0.4382*** (0.0259)	0.5233*** (0.0266)	0.3172 (0.2062)	0.3833*** (0.0319)
Observations	113,422	25,437	1,417	85,453
Fixed effects: origin × destination × product, origin × destination × year				
Number of foreign MNEs exporting the product to the destination (ln)	0.4506*** (0.0124)	0.4625*** (0.0188)	0.2479 (0.1778)	0.4435*** (0.0148)
Number of domestic MNEs exporting the product to the destination (ln)	0.2214*** (0.0151)	0.2831*** (0.0234)	0.3216* (0.1639)	0.1904*** (0.0175)
Observations	98,223	22,002	1,027	73,318
<i>Panel B. Dependent variable: ln exports of the product from the origin to the destination in the year; local (non-MNE) firms only</i>				
Fixed effects: origin × year, destination × year, product				
Number of foreign MNEs exporting the product to the destination (ln)	1.1331*** (0.0375)	1.2191*** (0.0363)	0.9420*** (0.2641)	1.0546*** (0.0493)
Number of domestic MNEs exporting the product to the destination (ln)	0.4707*** (0.0341)	0.5676*** (0.0339)	-0.6594 (0.4830)	0.4194*** (0.0419)
Observations	93,010	22,503	995	68,534
Fixed effects: origin × destination × product, origin × destination × year				
Number of foreign MNEs exporting the product to the destination (ln)	0.2413*** (0.0217)	0.3602*** (0.0292)	-0.1512 (0.3320)	0.1865*** (0.0250)
Number of domestic MNEs exporting the product to the destination (ln)	0.1250*** (0.0218)	0.2461*** (0.0341)	-0.7273 (0.6536)	0.0742*** (0.0238)
Observations	81,563	19,535	726	59,772

Fact 5 by sector, all goods and within each sector. Intensive margin: the coefficients of ln(number of foreign MNEs) and ln(number of domestic MNEs) exporting the product from the origin to the destination in the year (columns (3) and (4) of the full table in each scope). Observations with at least one foreign and one domestic MNE (plain logs). One observation = origin × destination × HS6 product × year, 2006–2022. A blank entry was not estimated (too few observations). SE clustered at origin–destination in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table A.10: Fact 5 by sector, extensive margin: presence of foreign and of domestic MNEs exporting the product to the destination, all goods and the three sectors

Sample	All goods	Agriculture	Mining & fuels	Manufacturing
<i>Panel A. Dependent variable: ln exports of the product from the origin to the destination in the year, all firms</i>				
Fixed effects: origin $\times$ year, destination $\times$ year, product				
At least one foreign MNE exports the product to the destination (0/1)	1.4990*** (0.0339)	1.7972*** (0.0558)	1.4255*** (0.0892)	1.4231*** (0.0344)
At least one domestic MNE exports the product to the destination (0/1)	1.4670*** (0.0416)	1.6133*** (0.0577)	1.7356*** (0.1413)	1.3878*** (0.0466)
Observations	1,829,626	295,788	32,912	1,496,381
Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year				
At least one foreign MNE exports the product to the destination (0/1)	0.6986*** (0.0158)	0.6929*** (0.0213)	0.7311*** (0.0560)	0.7021*** (0.0178)
At least one domestic MNE exports the product to the destination (0/1)	0.4917*** (0.0165)	0.5683*** (0.0237)	0.5718*** (0.1044)	0.4684*** (0.0186)
Observations	1,633,319	268,332	27,664	1,329,815
<i>Panel B. Dependent variable: ln exports of the product from the origin to the destination in the year, local (non-MNE) firms only</i>				
Fixed effects: origin $\times$ year, destination $\times$ year, product				
At least one foreign MNE exports the product to the destination (0/1)	1.2983*** (0.0346)	1.7649*** (0.0546)	1.5696*** (0.1099)	1.1402*** (0.0344)
At least one domestic MNE exports the product to the destination (0/1)	1.1316*** (0.0428)	1.3082*** (0.0539)	1.1333*** (0.1497)	1.0491*** (0.0490)
Observations	1,297,111	232,617	22,836	1,038,206
Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year				
At least one foreign MNE exports the product to the destination (0/1)	0.1356*** (0.0106)	0.2854*** (0.0179)	0.1991*** (0.0547)	0.0997*** (0.0114)
At least one domestic MNE exports the product to the destination (0/1)	0.1370*** (0.0138)	0.2610*** (0.0201)	0.2125*** (0.0899)	0.0984*** (0.0167)
Observations	1,137,550	208,479	18,666	904,114

Fact 5 by sector, all goods and within each sector. Extensive margin: the coefficients of the indicators for at least one foreign and at least one domestic MNE exporting the product from the origin to the destination in the year (columns (3) and (4) of the full table in each scope); all observations. One observation = origin $\times$ destination $\times$ HS6 product $\times$ year, 2006–2022. A blank entry was not estimated (too few observations). SE clustered at origin–destination in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

A.6 Fact 6

Table A.11: Fact 6 by sector: distance elasticity and its attenuation for foreign and domestic MNEs

Dep. var.: ln firm exports	All goods	Agriculture	Mining & fuels	Manufacturing
Origin, destination, year and product FE (column (1) of the full table)				
ln distance	-0.1645*** (0.0454)	-0.2135*** (0.0748)	-0.0926 (0.0819)	-0.1510*** (0.0454)
ln distance $\times$ foreign MNE	0.0487*** (0.0041)	0.0366*** (0.0034)	0.0566*** (0.0086)	0.0515*** (0.0048)
ln distance $\times$ domestic MNE	0.0318*** (0.0042)	0.0187*** (0.0046)	0.0706*** (0.0180)	0.0388*** (0.0052)
Observations	5,081,547	1,045,009	79,938	3,917,960
Origin $\times$ year, destination $\times$ year and product FE (column (2))				
ln distance	-0.1853*** (0.0460)	-0.2285*** (0.0711)	-0.1001 (0.0828)	-0.1742*** (0.0459)
ln distance $\times$ foreign MNE	0.0483*** (0.0040)	0.0358*** (0.0034)	0.0557*** (0.0087)	0.0512*** (0.0048)
ln distance $\times$ domestic MNE	0.0318*** (0.0042)	0.0182*** (0.0046)	0.0673*** (0.0183)	0.0390*** (0.0052)
Observations	5,081,360	1,044,787	79,612	3,917,765

Fact 6 in one table: the distance elasticity of firm exports and its attenuation for foreign and for domestic multinationals (columns (1) and (2) of the full table in each scope), all goods and within each sector. Firm $\times$ origin $\times$ destination $\times$ product $\times$ year; non-MNE flows are the omitted base, so the interaction is the difference from the base elasticity. SE clustered at origin–destination in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

A.7 Products

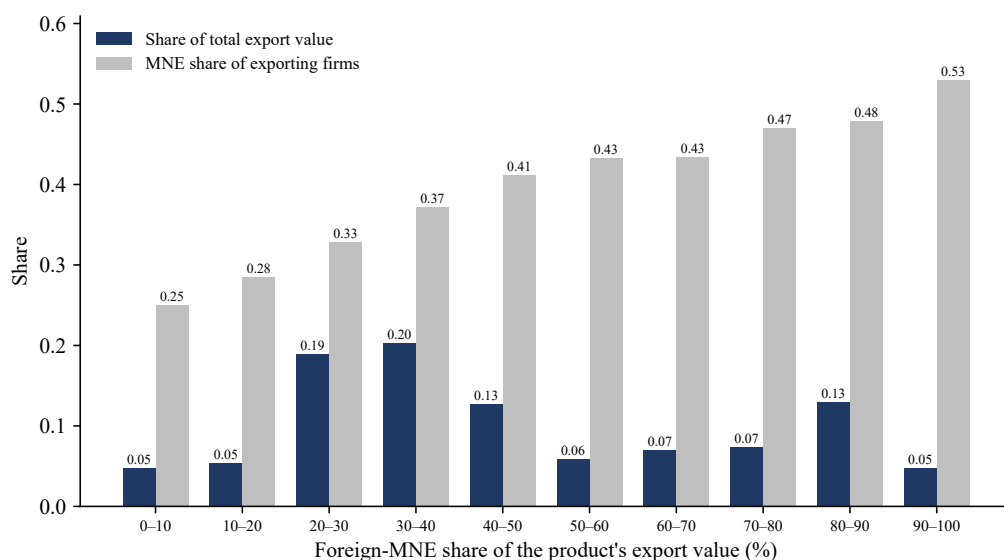


Figure A.9: Distribution of the HS6 foreign-MNE share: share of export value and MNE share of exporting firms, by bin.

Table A.12: Numbers behind the previous figure

Foreign share bin (%)	Share of value	MNE share of exporting firms	N HS6
0-10	0.048	0.250	1,426
10-20	0.054	0.285	731
20-30	0.189	0.329	652
30-40	0.202	0.372	591
40-50	0.127	0.411	519
50-60	0.059	0.433	442
60-70	0.070	0.435	386
70-80	0.074	0.470	311
80-90	0.129	0.479	311
90-100	0.048	0.530	560

MNE share of exporting firms = matched firm-cells (firm x destination x HS6 x year) over all firm-cells in the bin's products.